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(54) **ARTIFICIAL INTELLIGENCE FOR FLOOD MONITORING AND INSURANCE CLAIM FILING**

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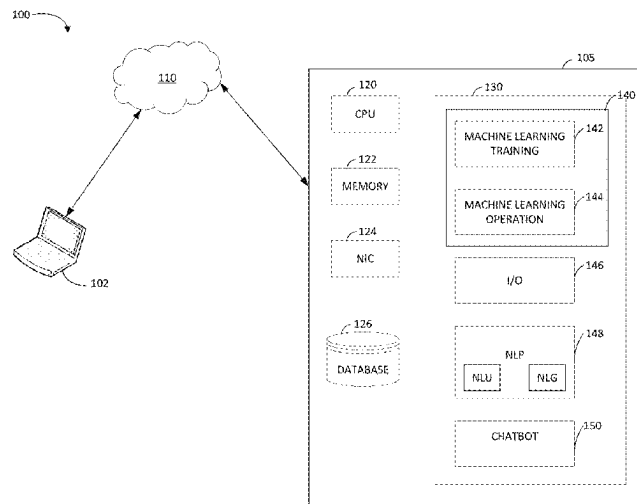
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(57) **ABSTRACT**

A computer system for flood monitoring and insurance provider notification, the computer system may include one or more processors configured to: detect a flood event in a structure, transmit information associated with the structure and a prompt for flood reimbursement services to a machine learning (ML) chatbot to cause the ML chatbot to file a flood reimbursement claim with an insurance provider having an insurance policy associated with the structure via telephone by converting a text output into a voice output.

20 Claims, 7 Drawing Sheets



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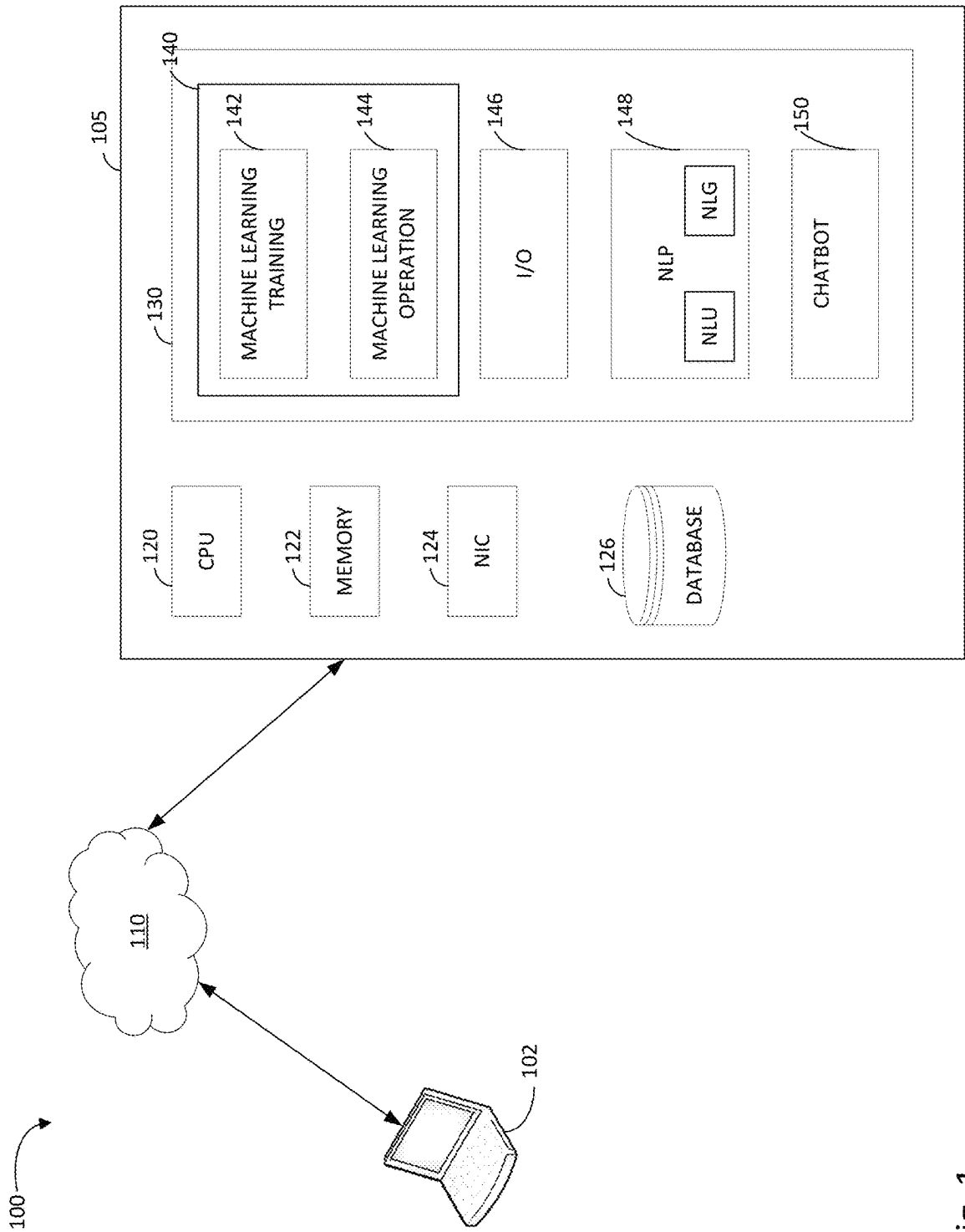


Fig. 1

Fig. 2A

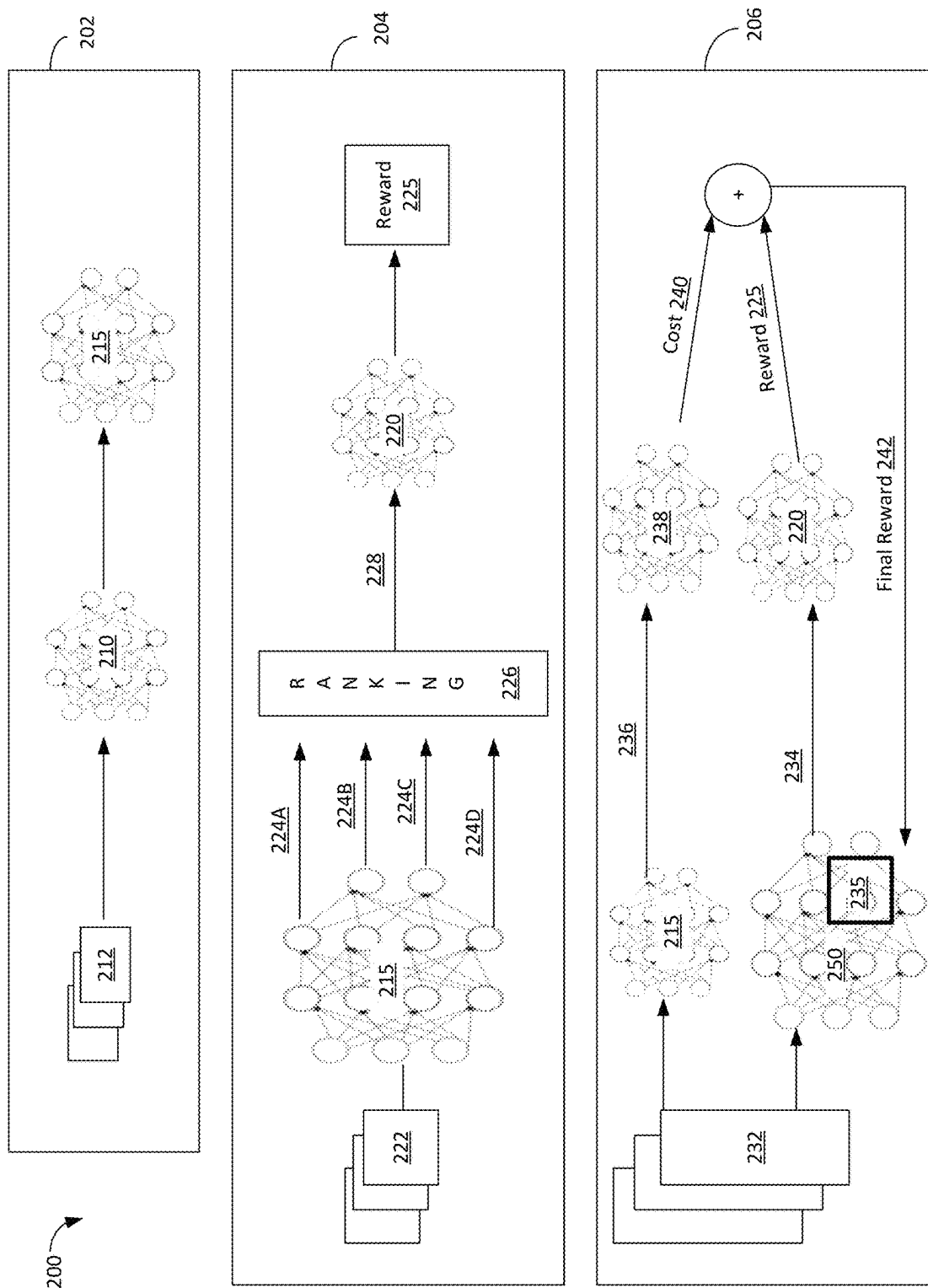
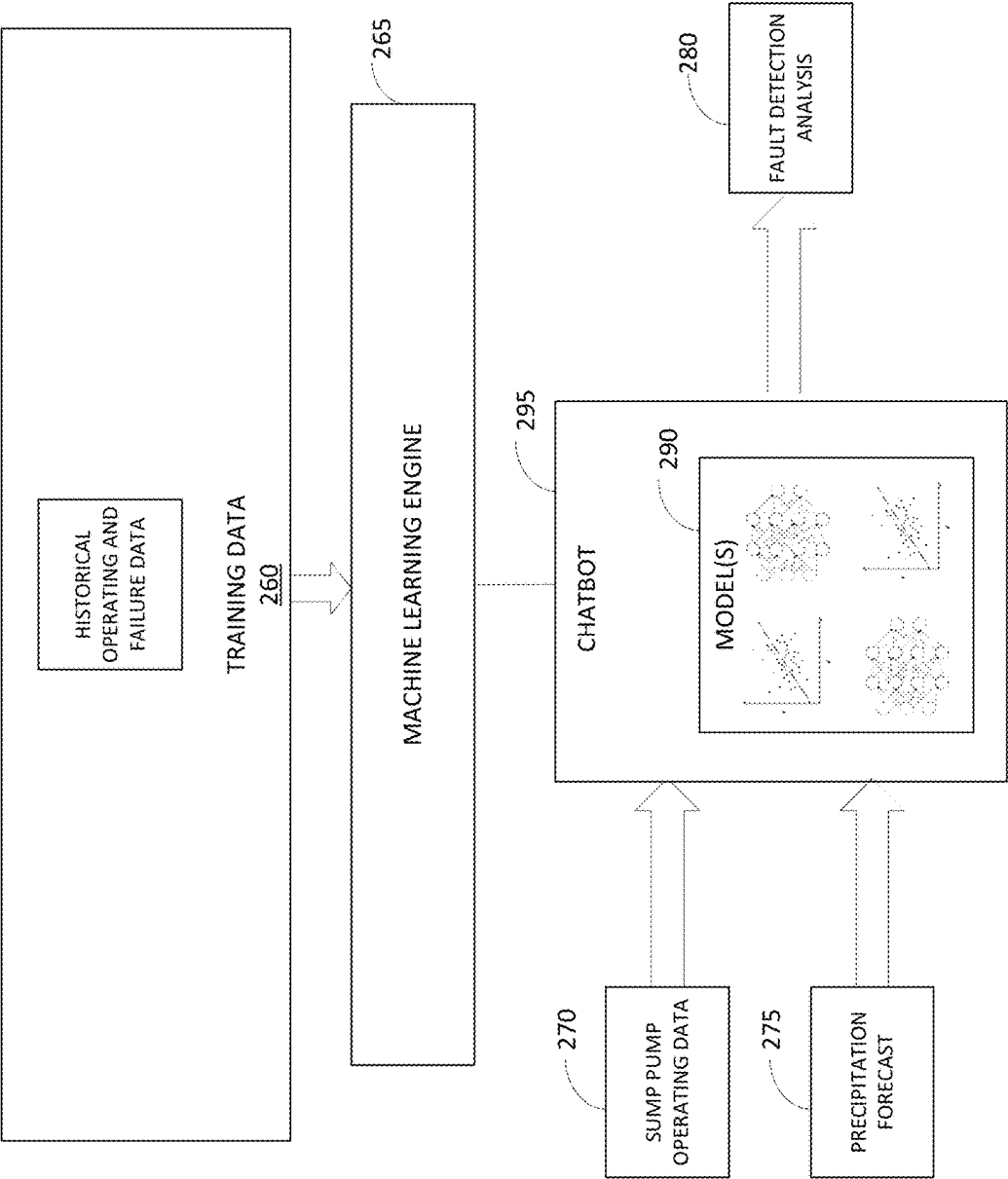


Fig. 2B



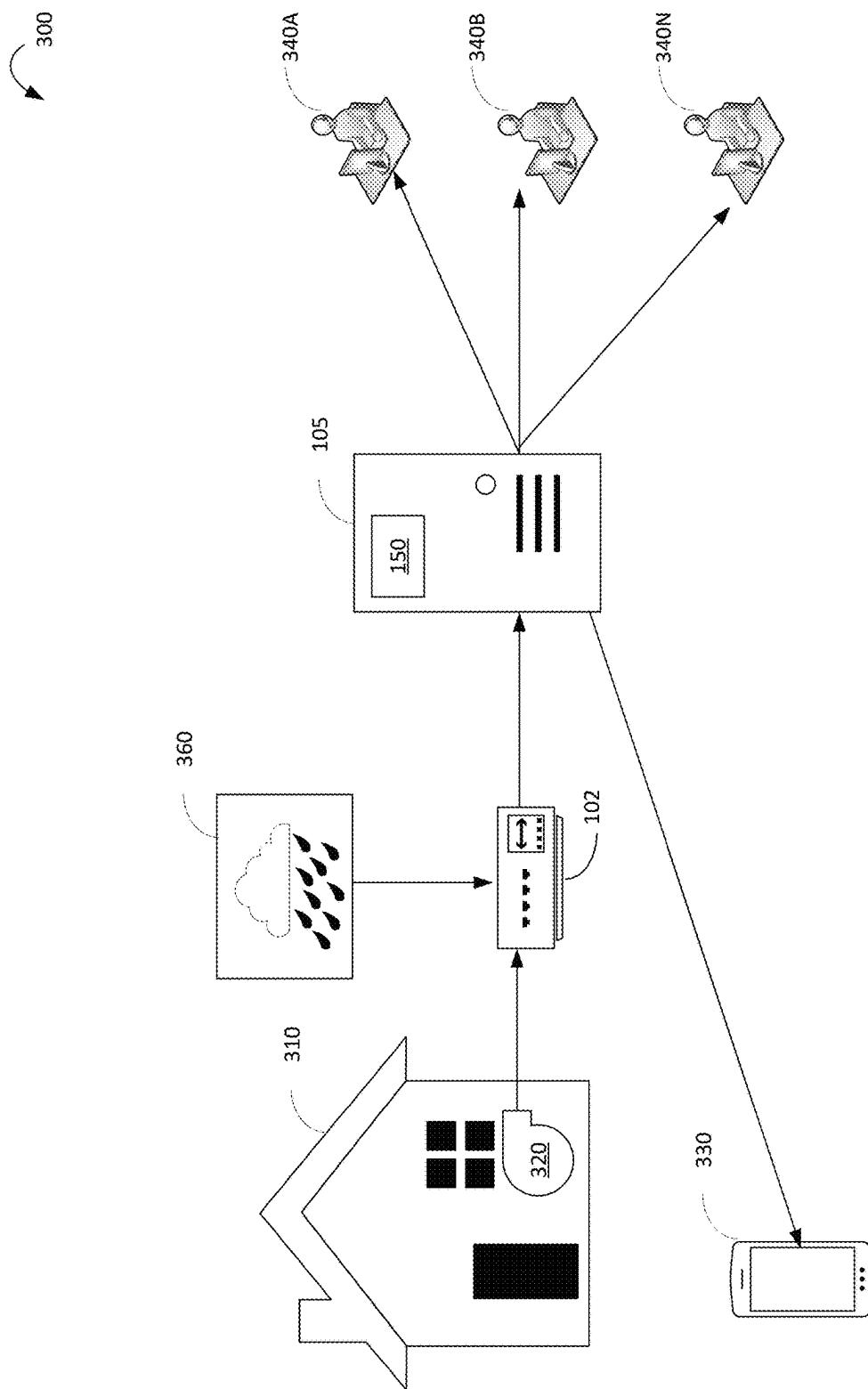


Fig. 3

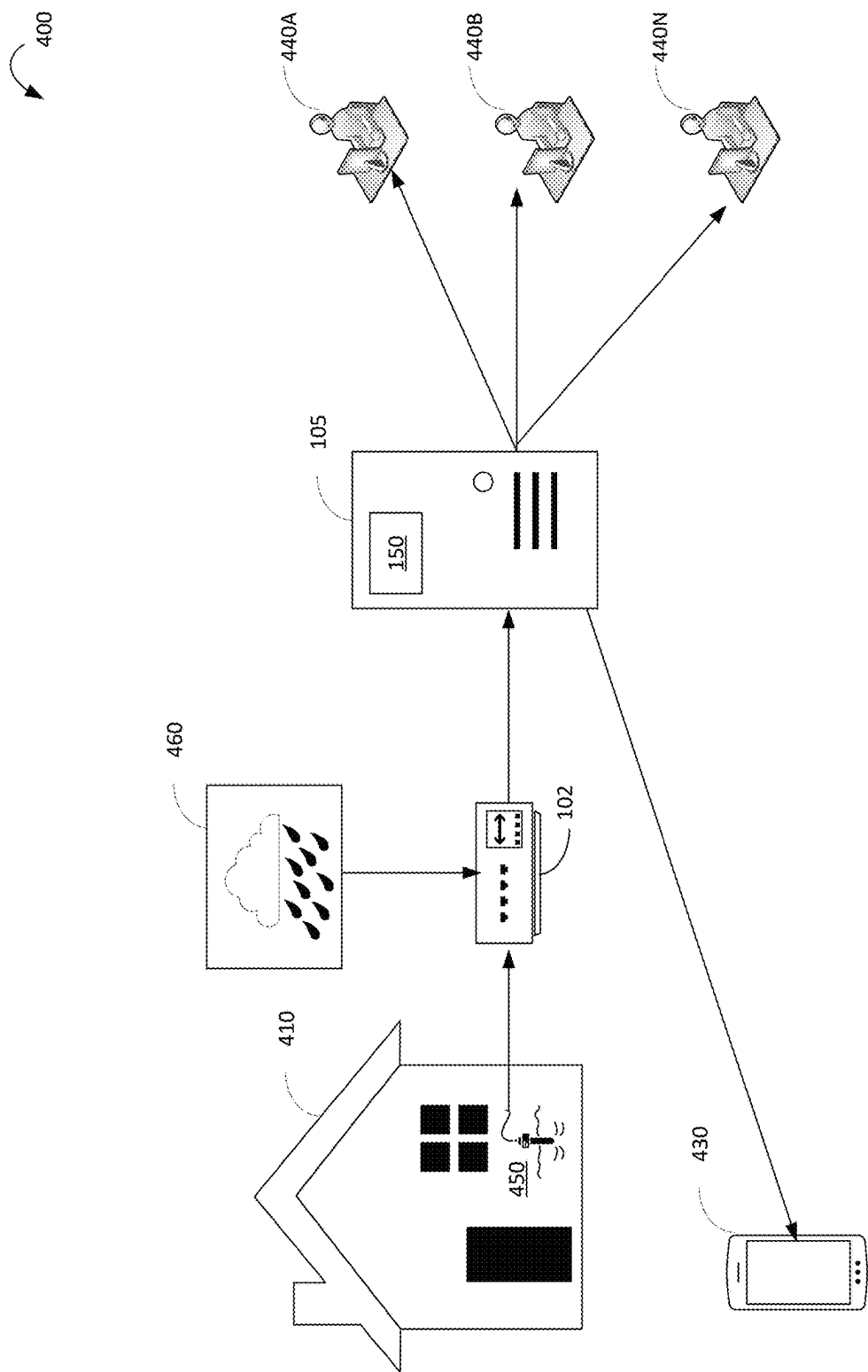


Fig. 4

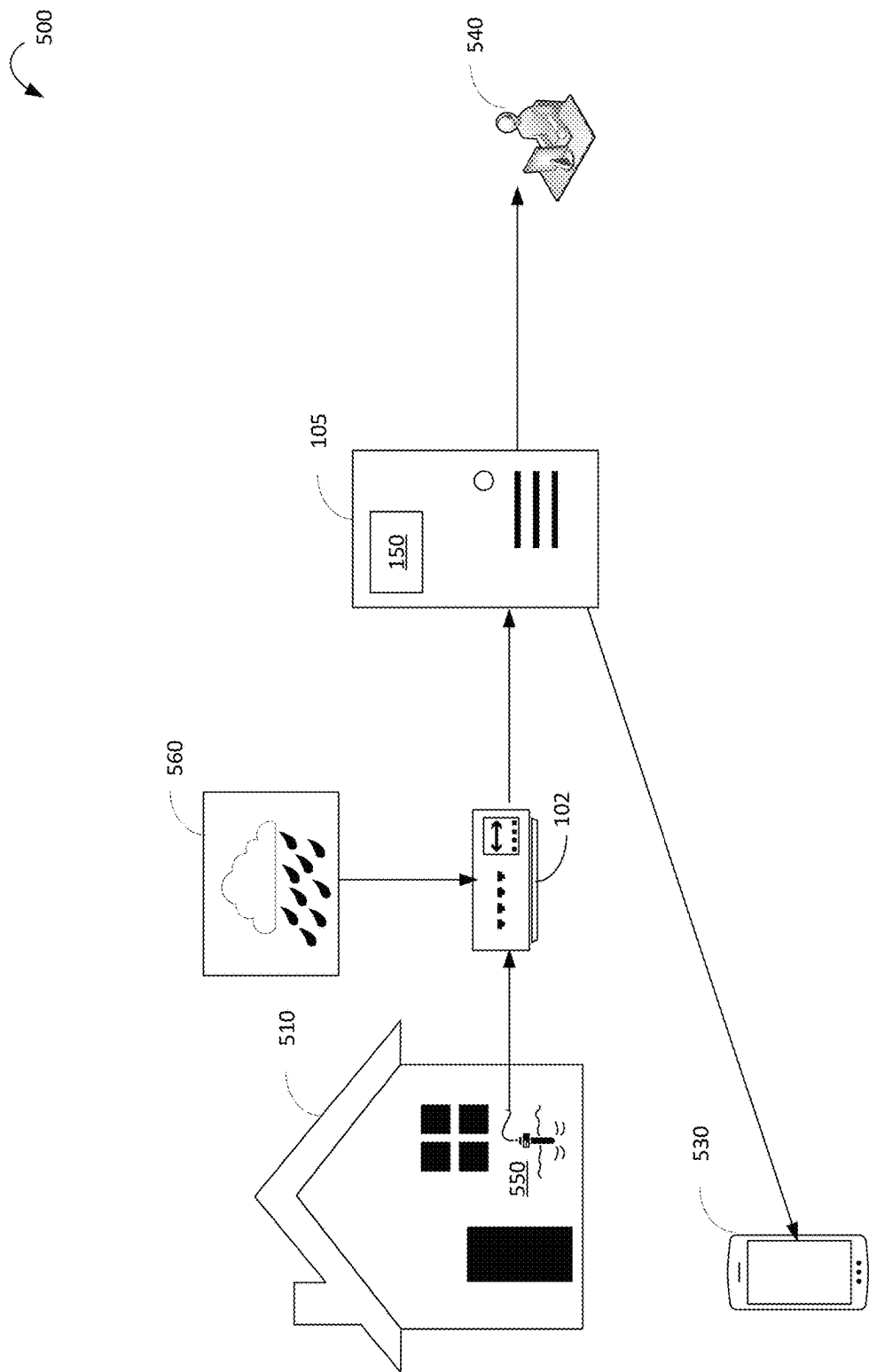


Fig. 5

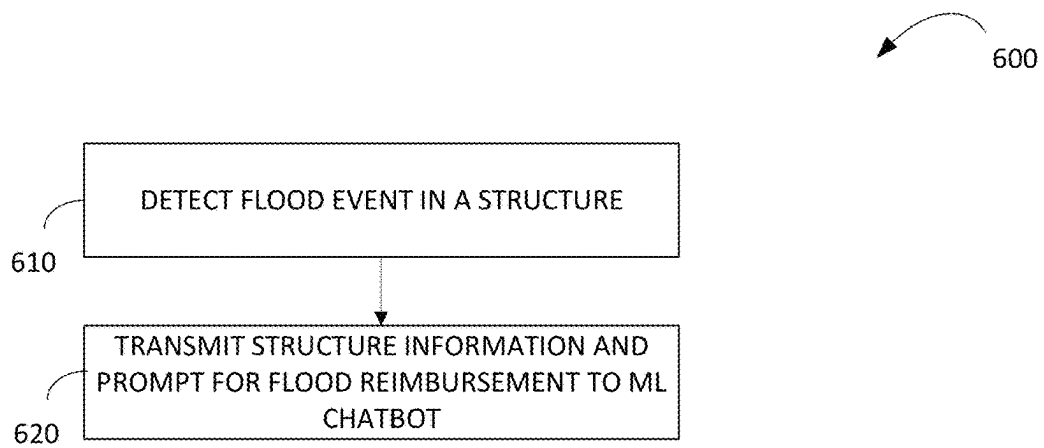


FIG. 6

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ARTIFICIAL INTELLIGENCE FOR FLOOD MONITORING AND INSURANCE CLAIM FILING

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to and the benefit of the filing date of provisional U.S. Patent Application No. 63/456,727 entitled "PARAMETRIC INSURANCE FOR WATER FLOW SENSORS FOR SUMP PUMPS," filed on Apr. 3, 2023, and provisional U.S. Patent Application No. 63/463,399 entitled "PARAMETRIC INSURANCE FOR WATER FLOW SENSORS FOR SUMP PUMPS," filed on May 2, 2023, the entire contents of both applications is hereby expressly incorporated herein by reference.

FIELD OF THE INVENTION

The present disclosure generally relates to sump pump and flood monitoring, and more particularly, remediation provider and/or insurance provider notification via a machine learning chatbot.

BACKGROUND

A sump pump may operate to prevent basements and other underground portions of a structure from flooding. Conventional sensors may detect that a sump pump is faulty and communicate an alert to a user associated with the structure. The user may then contact one or more service providers to have the sump pump repaired or replaced.

Conventional water sensors may detect a flood event in a structure and communicate an alert to a user associated with the structure. The user may then contact one or more service providers to request flood cleanup. The user may also contact an insurance provider to initiate a claim for flood damage.

The conventional sump pump and/or flood detection and service provider and/or insurance provider notification techniques may include additional shortcomings, inefficiencies, encumbrances, ineffectiveness, and/or other drawbacks.

SUMMARY

The present embodiments may relate to, inter alia, systems and methods for detecting flooding and automatically notifying insurance providers using a machine learning (ML) and/or artificial intelligence (AI) chatbot (or voice bot).

In one aspect, a computer-implemented method for flood monitoring and insurance provider notification using an ML chatbot (or voice bot) may be provided. The computer-implemented method may be implemented via one or more local or remote processors, servers, transceivers, sensors, memory units, mobile devices, wearables, smart watches, smart contact lenses, smart glasses, augmented reality glasses, virtual reality headsets, mixed or extended reality glasses or headsets, voice bots or chatbots, ChatGPT bots, and/or other electronic or electrical components, which may be in wired or wireless communication with one another. For example, in one instance, the computer-implemented method may include: (1) detecting, by one or more processors, a flood event in a structure; (2) transmitting, by the one or more processors, information associated with the structure and a prompt for flood reimbursement services to an ML chatbot; and/or (3) filing, by the one or more processors via

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the ML chatbot, a flood reimbursement claim with an insurance provider having an insurance policy associated with the structure via telephone by converting a text output into a voice output. The method may include additional, less, or alternate functionality or actions, including those discussed elsewhere herein.

In another aspect, a computer system for flood monitoring and insurance provider notification using an ML chatbot (or voice bot) may be provided. The computer system may include one or more local or remote processors, servers, transceivers, sensors, memory units, mobile devices, wearables, smart watches, smart contact lenses, smart glasses, augmented reality glasses, virtual reality headsets, mixed or extended reality glasses or headsets, voice bots, chatbots, ChatGPT bots, and/or other electronic or electrical components, which may be in wired or wireless communication with one another. For example, in one instance, the computer system may include one or more processors configured to: (1) detect a flood event in a structure; (2) transmit information associated with the structure and a prompt for flood reimbursement services to an ML chatbot; and/or (3) file, by the ML chatbot, a flood reimbursement claim with an insurance provider having an insurance policy associated with the structure via telephone by converting a text output into a voice output. The computer system may include additional, less, or alternate functionality, including that discussed elsewhere herein.

In another aspect, a non-transitory computer-readable medium storing processor-executable instructions for flood monitoring and insurance provider notification using an ML chatbot (or voice bot) may be provided. For example, in one instance, the computer-readable medium may include instructions that, when executed one or more processors to, cause the one or more processors to: (1) detect a flood event in a structure; (2) transmit information associated with the structure and a prompt for flood reimbursement services to an ML chatbot; and/or (3) file, by the ML chatbot, a flood reimbursement claim with an insurance provider having an insurance policy associated with the structure via telephone by converting a text output into a voice output. The instructions may direct additional, less, or alternate functionality, including that discussed elsewhere herein.

In one aspect, a computer-implemented method for flood monitoring and insurance provider notification using an AI chatbot (or voice bot) may be provided. The computer-implemented method may be implemented via one or more local or remote processors, servers, transceivers, sensors, memory units, mobile devices, wearables, smart watches, smart contact lenses, smart glasses, augmented reality glasses, virtual reality headsets, mixed or extended reality glasses or headsets, voice bots or chatbots, ChatGPT bots, and/or other electronic or electrical components, which may be in wired or wireless communication with one another. For example, in one instance, the computer-implemented method may include: (1) detecting, by one or more processors, a flood event in a structure; (2) transmitting, by the one or more processors, information associated with the structure and a prompt for flood reimbursement services to an AI chatbot; and/or (3) filing, by the one or more processors via the AI chatbot, a flood reimbursement claim with an insurance provider having an insurance policy associated with the structure via telephone by converting a text output into a voice output. The method may include additional, less, or alternate functionality or actions, including those discussed elsewhere herein.

In another aspect, a computer system for flood monitoring and insurance provider notification using an AI chatbot (or

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voice bot) may be provided. The computer system may include one or more local or remote processors, servers, transceivers, sensors, memory units, mobile devices, wearables, smart watches, smart contact lenses, smart glasses, augmented reality glasses, virtual reality headsets, mixed or extended reality glasses or headsets, voice bots, chatbots, ChatGPT bots, and/or other electronic or electrical components, which may be in wired or wireless communication with one another. For example, in one instance, the computer system may include one or more processors configured to: (1) detect a flood event in a structure; (2) transmit information associated with the structure and a prompt for flood reimbursement services to an AI chatbot; and/or (3) file, by the AI chatbot, a flood reimbursement claim with an insurance provider having an insurance policy associated with the structure via telephone by converting a text output into a voice output. The computer system may include additional, less, or alternate functionality, including that discussed elsewhere herein.

In another aspect, a non-transitory computer-readable medium storing processor-executable instructions for flood monitoring and insurance provider notification using an AI chatbot (or voice bot) may be provided. For example, in one instance, the computer-readable medium may include instructions that, when executed one or more processors to, cause the one or more processors to: (1) detect a flood event in a structure; (2) transmit information associated with the structure and a prompt for flood reimbursement services to an AI chatbot; and/or (3) file, by the AI chatbot, a flood reimbursement claim with an insurance provider having an insurance policy associated with the structure via telephone by converting a text output into a voice output. The instructions may direct additional, less, or alternate functionality, including that discussed elsewhere herein.

Additional, alternate and/or fewer actions, steps, features and/or functionality may be included in an aspect and/or embodiments, including those described elsewhere herein.

BRIEF DESCRIPTION OF THE DRAWINGS

The figures described below depict various aspects of the applications, methods, and systems disclosed herein. It should be understood that each figure depicts one embodiment of a particular aspect of the disclosed applications, systems and methods, and that each of the figures is intended to accord with a possible embodiment thereof. Furthermore, wherever possible, the following description refers to the reference numerals included in the following figures, in which features depicted in multiple figures are designated with consistent reference numerals.

FIG. 1 depicts a block diagram of an exemplary computer system in which methods and systems for sump pump and/or flood monitoring and service provider and/or insurance provider notification are implemented.

FIG. 2A depicts a combined block and logic diagram in which exemplary computer-implemented methods and systems for training an ML chatbot are implemented.

FIG. 2B depicts a combined block and logic diagram of an exemplary server generating sump pump fault detection analysis using generative AI/ML.

FIG. 3 depicts an exemplary environment for sump pump monitoring and service provider notification using an ML chatbot.

FIG. 4 depicts an exemplary environment for flood monitoring and service provider notification using an ML chatbot.

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FIG. 5 depicts an exemplary environment for flood monitoring and insurance provider notification using an ML chatbot.

FIG. 6 depicts an exemplary computer-implemented method for flood monitoring and insurance claim filing.

Advantages will become more apparent to those skilled in the art from the following description of the preferred embodiments which have been shown and described by way of illustration. As will be realized, the present embodiments may be capable of other and different embodiments, and their details are capable of modification in various respects. Accordingly, the drawings and description are to be regarded as illustrative in nature and not as restrictive.

DETAILED DESCRIPTION

Overview

The computer systems and methods disclosed herein generally relate to, inter alia, methods and systems for sump pump and flood monitoring, and more particularly, remediation provider and/or insurance provider notification using a machine learning (ML) and/or artificial intelligence (AI) chatbot and/or voice bot.

Some embodiments may include one or more of: (1) sump pump fault detection and automatic replacement or repair service requests; (2) flood detection and automatic flood remediation requests; and (3) flood detection and automatic insurance claim filing.

Exemplary Computing Environment

FIG. 1 depicts a block diagram of an exemplary computing environment 100 in which sump pump and/or flood monitoring and remediation provider and/or insurance provider notification may be performed, in accordance with various aspects discussed herein.

As illustrated, the computing environment 100 includes a monitoring device 102. The computing environment 100 may further include an electronic network 110 communicatively coupling other aspects of the computing environment 100.

The monitoring device 102 may be any suitable device and include one or more Internet of Things (IoT) hubs, smart home devices, mobile devices, wearables, smart watches, smart contact lenses, smart glasses, AR glasses/headsets, virtual reality (VR) glasses/headsets, mixed or extended reality glasses/headsets, voice bots or chatbots, ChatGPT bots, displays, display screens, visuals, and/or other electronic or electrical component. The monitoring device 102 may include a memory and a processor for, respectively, storing and executing one or more modules. The memory may include one or more suitable storage media such as a magnetic storage device, a solid-state drive, random access memory (RAM), etc. The monitoring device 102 may access services or other components of the computing environment 100 via the network 110.

As described herein and in an aspect, one or more servers 105 may perform the functionalities as part of a cloud network or may otherwise communicate with other hardware or software components within one or more cloud computing environments to send, retrieve, or otherwise analyze data or information described herein. For example, in certain aspects of the present techniques, the computing environment 100 may comprise an on-premise computing environment, a multi-cloud computing environment, a public cloud computing environment, a private cloud computing environment, and/or a hybrid cloud computing environment. For example, an entity (e.g., a business) providing a chatbot to enable remediation provider and/or insurance provider

notification may host one or more services in a public cloud computing environment (e.g., Alibaba Cloud, Amazon Web Services (AWS), Google Cloud, IBM Cloud, Microsoft Azure, etc.). The public cloud computing environment may be a traditional off-premise cloud (i.e., not physically hosted at a location owned/controlled by the business). Alternatively, or in addition, aspects of the public cloud may be hosted on-premise at a location owned/controlled by the a structure owner or lessee. The public cloud may be partitioned using visualization and multi-tenancy techniques and may include one or more infrastructure-as-a-service (IaaS) and/or platform-as-a-service (PaaS) services.

The network **110** may comprise any suitable network or networks, including a local area network (LAN), wide area network (WAN), Internet, or combination thereof. For example, the network **110** may include a wireless cellular service (e.g., 4G, 5G, 6G, etc.). Generally, the network **110** enables bidirectional communication between the monitoring device **102** and the servers **105**. In one aspect, the network **110** may comprise a cellular base station, such as cell tower(s), communicating to the one or more components of the computing environment **100** via wired/wireless communications based on any one or more of various mobile phone standards, including NMT, GSM, CDMA, UMTS, LTE, 5G, 6G, or the like. Additionally or alternatively, the network **110** may comprise one or more routers, wireless switches, or other such wireless connection points communicating to the components of the computing environment **100** via wireless communications based on any one or more of various wireless standards, including by non-limiting example, IEEE 802.11a/b/g/n/ac/ax/be (WIFI), Bluetooth, and/or the like.

The processor **120** may include one or more suitable processors (e.g., central processing units (CPUs) and/or graphics processing units (GPUS)). The processor **120** may be connected to the memory **122** via a computer bus (not depicted) responsible for transmitting electronic data, data packets, or otherwise electronic signals to and from the processor **120** and memory **122** in order to implement or perform the machine-readable instructions, methods, processes, elements or limitations, as illustrated, depicted, or described for the various flowcharts, illustrations, diagrams, figures, and/or other disclosure herein. The processor **120** may interface with the memory **122** via a computer bus to execute an operating system (OS) and/or computing instructions contained therein, and/or to access other services/aspects. For example, the processor **120** may interface with the memory **122** via the computer bus to create, read, update, delete, or otherwise access or interact with the data stored in the memory **122** and/or a database **126**.

The memory **122** may include one or more forms of volatile and/or non-volatile, fixed and/or removable memory, such as read-only memory (ROM), electronic programmable read-only memory (EPROM), random access memory (RAM), erasable electronic programmable read-only memory (EEPROM), and/or other hard drives, flash memory, MicroSD cards, and others. The memory **122** may store an operating system (OS) (e.g., Microsoft Windows, Linux, UNIX, MacOS, etc.) capable of facilitating the functionalities, apps, methods, or other software as discussed herein.

The memory **122** may store a plurality of computing modules **130**, implemented as respective sets of computer-executable instructions (e.g., one or more source code libraries, trained ML models such as neural networks, convolutional neural networks, etc.) as described herein.

In general, a computer program or computer based product, application, or code (e.g., the model(s), such as ML models, or other computing instructions described herein) may be stored on a computer usable storage medium, or tangible, non-transitory computer-readable medium (e.g., standard random access memory (RAM), an optical disc, a universal serial bus (USB) drive, or the like) having such computer-readable program code or computer instructions embodied therein, wherein the computer-readable program code or computer instructions may be installed on or otherwise adapted to be executed by the processor(s) **120** (e.g., working in connection with the respective operating system in memory **122**) to facilitate, implement, or perform the machine readable instructions, methods, processes, elements or limitations, as illustrated, depicted, or described for the various flowcharts, illustrations, diagrams, figures, and/or other disclosure herein. In this regard, the program code may be implemented in any desired program language, and may be implemented as machine code, assembly code, byte code, interpretable source code or the like (e.g., via Golang, Python, C, C++, C#, Objective-C, Java, Scala, ActionScript, JavaScript, HTML, CSS, XML, etc.).

The database **126** may be a relational database, such as Oracle, DB2, MySQL, a NoSQL based database, such as MongoDB, or another suitable database. The database **126** may store data and be used to train and/or operate one or more ML models, chatbots, and/or voice bots.

In one aspect, the computing modules **130** may include an ML module **140**. The ML module **140** may include ML training module (MLTM) **142** and/or ML operation module (MLOM) **144**. In some embodiments, at least one of a plurality of ML methods and algorithms may be applied by the ML module **140**, which may include, but are not limited to: linear or logistic regression, instance-based algorithms, regularization algorithms, decision trees, Bayesian networks, cluster analysis, association rule learning, artificial neural networks, deep learning, combined learning, reinforced learning, dimensionality reduction, and support vector machines. In various embodiments, the implemented ML methods and algorithms are directed toward at least one of a plurality of categorizations of ML, such as supervised learning, unsupervised learning, and reinforcement learning.

In one aspect, the ML based algorithms may be included as a library or package executed on server(s) **105**. For example, libraries may include the TensorFlow based library, the PyTorch library, the HuggingFace library, and/or the scikit-learn Python library.

In one embodiment, the ML module **140** employs supervised learning, which involves identifying patterns in existing data to make predictions about subsequently received data. Specifically, the ML module is "trained" (e.g., via MLTM **142**) using training data, which includes example inputs and associated example outputs. Based upon the training data, the ML module **140** may generate a predictive function which maps outputs to inputs and may utilize the predictive function to generate ML outputs based upon data inputs. The exemplary inputs and exemplary outputs of the training data may include any of the data inputs or ML outputs described above. In the exemplary embodiments, a processing element may be trained by providing it with a large sample of data with known characteristics or features.

In another embodiment, the ML module **140** may employ unsupervised learning, which involves finding meaningful relationships in unorganized data. Unlike supervised learning, unsupervised learning does not involve user-initiated training based upon example inputs with associated outputs. Rather, in unsupervised learning, the ML module **140** may

organize unlabeled data according to a relationship determined by at least one ML method/algorithm employed by the ML module **140**. Unorganized data may include any combination of data inputs and/or ML outputs as described above.

In yet another embodiment, the ML module **140** may employ reinforcement learning, which involves optimizing outputs based upon feedback from a reward signal. Specifically, the ML module **140** may receive a user-defined reward signal definition, receive a data input, utilize a decision-making model to generate the ML output based upon the data input, receive a reward signal based upon the reward signal definition and the ML output, and alter the decision-making model so as to receive a stronger reward signal for subsequently generated ML outputs. Other types of ML may also be employed, including deep or combined learning techniques.

The MLTM **142** may receive labeled data at an input layer of a model having a networked layer architecture (e.g., an artificial neural network, a convolutional neural network, etc.) for training the one or more ML models. The received data may be propagated through one or more connected deep layers of the ML model to establish weights of one or more nodes, or neurons, of the respective layers. Initially, the weights may be initialized to random values, and one or more suitable activation functions may be chosen for the training process. The present techniques may include training a respective output layer of the one or more ML models. The output layer may be trained to output a prediction, for example.

The MLOM **144** may comprising a set of computer-executable instructions implementing ML loading, configuration, initialization and/or operation functionality. The MLOM **144** may include instructions for storing trained models (e.g., in the electronic database **126**). As discussed, once trained, the one or more trained ML models may be operated in inference mode, whereupon when provided with de novo input that the model has not previously been provided, the model may output one or more predictions, classifications, etc., as described herein.

In one aspect, the computing modules **130** may include an input/output (I/O) module **146**, comprising a set of computer-executable instructions implementing communication functions. The I/O module **146** may include a communication component configured to communicate (e.g., send and receive) data via one or more external/network port(s) to one or more networks or local terminals, such as the computer network **110** and/or the user device **102** (for rendering or visualizing) described herein. In one aspect, the servers **105** may include a client-server platform technology such as ASP.NET, Java J2EE, Ruby on Rails, Node.js, a web service or online API, responsive for receiving and responding to electronic requests.

I/O module **146** may further include or implement an operator interface configured to present information to an administrator or operator and/or receive inputs from the administrator and/or operator. An operator interface may provide a display screen. The I/O module **146** may facilitate I/O components (e.g., ports, capacitive or resistive touch sensitive input panels, keys, buttons, lights, LEDs), which may be directly accessible via, or attached to, servers **105** or may be indirectly accessible via or attached to the monitoring device **102**. According to an aspect, an administrator or operator may access the servers **105** via the monitoring device **102** to review information, make changes, input training data, initiate training via the MLTM **142**, and/or

perform other functions (e.g., operation of one or more trained models via the MLOM **144**).

In one aspect, the computing modules **130** may include one or more NLP modules **148** comprising a set of computer-executable instructions implementing NLP, natural language understanding (NLU) and/or natural language generator (NLG) functionality. The NLP module **148** may be responsible for transforming the user input (e.g., unstructured conversational input such as speech or text) to an interpretable format. The NLP module **148** may include an NLU to understand the intended meaning of utterances and/or prompts, among other things. The NLP module **148** may include an NLG, which may provide text summarization, machine translation, and/or dialog where structured data is transformed into natural conversational language (i.e., unstructured) for output to the user.

In one aspect, the computing modules **130** may include one or more chatbots and/or voice bots **150** which may be programmed to simulate human conversation, interact with users, understand their needs, and recommend an appropriate line of action with minimal and/or no human intervention, among other things. This may include providing the best response of any query that it receives and/or asking follow-up questions.

In some embodiments, the voice bots or chatbots **150** discussed herein may be configured to utilize AI and/or ML techniques. For instance, the voice bot or chatbot **150** may be a ChatGPT chatbot. The voice bot or chatbot **150** may employ supervised or unsupervised ML techniques, which may be followed by, and/or used in conjunction with, reinforced or reinforcement learning techniques. The voice bot or chatbot **150** may employ the techniques utilized for ChatGPT.

Noted above, in some embodiments, a chatbot **150** or other computing device may be configured to implement ML, such that the server **105** "learns" to analyze, organize, and/or process data without being explicitly programmed. ML may be implemented through the ML methods and algorithms. In one exemplary embodiment, the ML module **140** may be configured to implement the ML.

For example, in an aspect, the server **105** may initiate a telephone session over the network **110** with a service provider or insurance agent, e.g., so the server **105** may request sump pump replacement or repair, flood remediation, and or flood reimbursement. As another example, the server **105** may initiate a text-based chat session over the network **110** with a service provider or insurance agent. The chatbot **150** may receive utterances and/or prompts from the service provider or insurance agent, i.e., the input from the provider or agent from which the chatbot **150** needs to derive intents from. The utterances and/or prompts may be processed using NLP module **148** and/or ML module **140** via one or more ML models to recognize what the provider or agent says or types, understand the meaning, determine the appropriate action, and/or respond with language the provider or agent can understand.

In one aspect, the server **105** may host and/or provide an application (e.g., a mobile application), and/or a website configured to provide the application, to receive sump pump and/or flood sensor data from the monitoring device **102**. In one aspect, the server **105** may store code in memory **122** which when executed by CPU **120** may provide the website and/or application. In a further aspect, the server **105** may receive the sump pump and/or flood sensor data from the monitoring device **102**. In some embodiments, the sump pump and/or flood sensor data may indicate a repository, file location, and/or other data store at which the source code

and/or privacy policy may be maintained. In some embodiments, the server **105** may store at least a portion of the sump pump and/or flood sensor data in the database **126**. The data stored in the database **126** may be cleaned, labeled, vectorized, weighted and/or otherwise processed, especially processing suitable for data used in any aspect of ML.

In one aspect, the server **105** may host and/or provide an application to initiate and conduct the telephone session and/or chat session with the service provider or insurance agent. In one aspect, the server **105** may store code in memory **122** which when executed by CPU **120** may provide the application. In another aspect, the server **105** may store the received utterances and/or prompts from the service provider or insurance agent, recognition of what the provider or agent says or types, understanding of the meaning, determination of the appropriate action, and/or response in the database **126**. The data may be cleaned, labeled, vectorized, weighted and/or otherwise processed, especially processing suitable for data used in any aspect of ML.

In one aspect, when the server **105** evaluates the sump pump data, flood sensor data, and/or service provider and/or insurance agent telephone and/or chat session data, the data may be stored in the database **126**. In one aspect, the server **105** may use the stored data to generate, train and/or retrain one or more ML models and/or chatbots **150**, and/or for any other suitable purpose.

In operation, ML model training module **142** may access database **126** or any other data source for training data suitable to generate one or more ML models appropriate to receive and/or process the sump pump data, flood sensor data, and/or service provider and/or insurance agent telephone and/or chat session data, e.g., a chatbot **150**. The training data may be sample data with assigned relevant and comprehensive labels (classes or tags) used to fit the parameters (weights) of an ML model with the goal of training it by example. In one aspect, training data may include historical data from past notices of sump pump faults and/or flood events. The historical data may include service provider names, cost estimates, schedule availabilities, as well as any other suitable training data. In one aspect, training data may include transcripts of telephone and/or chat sessions with service providers and/or insurance agents. The training data may include user ratings, e.g., **1** to **10** score, of the output provided by the ML model. The ML model trained on such training data will have an improved capability to successfully communicate with a service provider and/or insurance agent when compared to a conventional ML chatbot. In one aspect, once an appropriate ML model is trained and validated to provide accurate predictions and/or responses, e.g., the chatbot **150**, the trained model and/or chatbot **150** may be loaded into MLOM **144** at runtime, may process the service provider and/or insurance agent inputs, utterances, and/or prompts, and may generate as an output conversational dialog.

While various embodiments, examples, and/or aspects disclosed herein may include training and generating one or more ML models and/or chatbot **150** for the server **105** to load at runtime, it is also contemplated that one or more appropriately trained ML models and/or chatbot **150** may already exist (e.g., in database **126**) such that the server **105** may load an existing trained ML model and/or chatbot **150** at runtime. It is further contemplated that the server **105** may retrain, update and/or otherwise alter an existing ML model and/or chatbot **150** before loading the model at runtime.

Although the computing environment **100** is shown to include one monitoring device **102**, one server **105**, and one network **110**, it should be understood that different numbers

of monitoring devices **102**, networks **110**, and/or servers **105** may be utilized. In one example, the computing environment **100** may include a plurality of servers **105** and hundreds or thousands of monitoring devices **102**, all of which may be interconnected via the network **110**. Furthermore, the database storage or processing performed by the one or more servers **105** may be distributed among a plurality of servers **105** in an arrangement known as “cloud computing.” This configuration may provide various advantages, such as enabling near real-time uploads and downloads of information as well as periodic uploads and downloads of information.

The computing environment **100** may include additional, fewer, and/or alternate components, and may be configured to perform additional, fewer, or alternate actions, including components/actions described herein. Although the computing environment **100** is shown in FIG. **1** as including one instance of various components such as monitoring device **102**, server **105**, and network **110**, etc., various aspects include the computing environment **100** implementing any suitable number of any of the components shown in FIG. **1** and/or omitting any suitable ones of the components shown in FIG. **1**. For instance, information described as being stored at server database **126** may be stored at memory **122**, and thus database **126** may be omitted. Moreover, various aspects include the computing environment **100** including any suitable additional component(s) not shown in FIG. **1**, such as but not limited to the exemplary components described above. Furthermore, it should be appreciated that additional and/or alternative connections between components shown in FIG. **1** may be implemented. As just one example, server **105** and monitoring device **102** may be connected via a direct communication link (not shown in FIG. **1**) instead of, or in addition to, via network **130**.

Exemplary Training of the ML Chatbot Model

An enterprise may be able to use programmable chatbots, such as the chatbot **150** (e.g., ChatGPT), to provide tailored, conversational-like remediation, repair, and/or reimbursement requests. In one aspect, the chatbot may be capable of making a request, providing relevant information, answering service provider and/or insurance agent questions, any of which may assist and/or replace the need for human initiated conversations. Additionally, the chatbot may generate data from service provider and/or insurance agent interactions which the enterprise may use to personalize future support and/or improve the chatbot’s functionality, e.g., when retraining and/or fine-tuning the chatbot.

The ML chatbot may provide advanced features as compared to a non-ML chatbot, which may include and/or derive functionality from a large language model (LLM). The ML chatbot may be trained on a server, such as server **105**, using large training datasets of text which may provide sophisticated capability for natural-language tasks, such as answering questions and/or holding conversations. The ML chatbot may include a general-purpose pretrained LLM which, when provided with a starting set of words (prompt) as an input, may attempt to provide an output (response) of the most likely set of words that follow from the input. In one aspect, the prompt may be provided to, and/or the response received from, the ML chatbot and/or any other ML model, via a user interface of the server. This may include a user interface device operably connected to the server via an I/O module, such as the I/O module **146**. Exemplary user interface devices may include a touchscreen, a keyboard, a mouse, a microphone, a speaker, a display, and/or any other suitable user interface devices.

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Multi-turn (i.e., back-and-forth) conversations may require LLMs to maintain context and coherence across multiple service provider and/or insurance agent utterances and/or prompts, which may require the ML chatbot to keep track of an entire conversation history as well as the current state of the conversation. The ML chatbot may rely on various techniques to engage in conversations with service providers and/or insurance agents, which may include the use of short-term and long-term memory. Short-term memory may temporarily store information (e.g., in the memory 122 of the server 105) that may be required for immediate use and may keep track of the current state of the conversation and/or to understand the service provider's and/or insurance agent's latest statement to generate an appropriate response or question. Long-term memory may include persistent storage of information (e.g., on database 126 of the server 105) which may be accessed over an extended period of time. The long-term memory may be used by the ML chatbot to store information about the service provider and/or insurance agent (e.g., preferences, chat history, etc.) and may be useful for improving an overall more effective conversation by enabling the ML chatbot to personalize and/or provide more informed responses or questions.

The system and methods to generate and/or train an ML chatbot model (e.g., via the ML module 140 of the server 105) which may be used by the ML chatbot, may consist of three steps: (1) a supervised fine-tuning (SFT) step where a pretrained language model (e.g., an LLM) may be fine-tuned on a relatively small amount of demonstration data curated by human labelers to learn a supervised policy (SFT ML model) which may generate responses/outputs from a selected list of prompts/inputs. The SFT ML model may represent a cursory model for what may be later developed and/or configured as the ML chatbot model; (2) a reward model step where human labelers may rank numerous SFT ML model responses and/or questions to evaluate the responses and/or questions which best mimic preferred human responses and/or questions, thereby generating comparison data. The reward model may be trained on the comparison data; and/or (3) a policy optimization step in which the reward model may further fine-tune and improve the SFT ML model. The outcome of this step may be the ML chatbot model using an optimized policy. In one aspect, step one may take place only once, while steps two and three may be iterated continuously, e.g., more comparison data is collected on the current ML chatbot model, which may be used to optimize/update the reward model and/or further optimize/update the policy.

Supervised Fine-Tuning ML Model

FIG. 2A depicts a combined block and logic diagram 200 for training an ML chatbot model, in which the techniques described herein may be implemented, according to some embodiments. Some of the blocks in FIG. 2 may represent hardware and/or software components, other blocks may represent data structures or memory storing these data structures, registers, or state variables (e.g., 212), and other blocks may represent output data (e.g., 225). Input and/or output signals may be represented by arrows labeled with corresponding signal names and/or other identifiers. The methods and systems may include one or more servers 202, 204, 206, such as the server 105 of FIG. 1.

In one aspect, the server 202 may fine-tune a pretrained language model 210. The pretrained language model 210 may be obtained by the server 202 and be stored in a memory, such as memory 122 and/or database 126. The pretrained language model 210 may be loaded into an ML

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training module, such as MLTL 142, by the server 202 for retraining/fine-tuning. A supervised training dataset 212 may be used to fine-tune the pretrained language model 210 wherein each data input prompt to the pretrained language model 210 may have a known output response for the pretrained language model 210 to learn from. The supervised training dataset 212 may be stored in a memory of the server 202, e.g., the memory 122 or the database 126. In one aspect, the data labelers may create the supervised training dataset 212 prompts and appropriate responses. The pretrained language model 210 may be fine-tuned using the supervised training dataset 212 resulting in the SFT ML model 215 which may provide appropriate responses to service provider and/or insurance agent prompts once trained. The trained SFT ML model 215 may be stored in a memory of the server 202, e.g., memory 122 and/or database 126.

In one aspect, the supervised training dataset 212 may include prompts and responses which may be relevant to requesting sump pump repair and/or replacement, flood remediation services, and/or flood reimbursement. For example, a service provider and/or insurance agent prompt may include a question about the structure associated with the sump pump fault and/or flood event. Appropriate responses from the trained SFT ML model 215 may include providing the service provider an address, indication of residential vs. commercial use, size, number of stories, etc. about the structure.

Training the Reward Model

In one aspect, training the ML chatbot model 250 may include the server 204 training a reward model 220 to provide as an output a scalar value/reward 225. The reward model 220 may be required to leverage reinforcement learning with human feedback (RLHF) in which a model (e.g., ML chatbot model 250) learns to produce outputs which maximize its reward 225, and in doing so may provide responses which are better aligned to service provider and/or insurance agent prompts.

Training the reward model 220 may include the server 204 providing a single prompt 222 to the SFT ML model 215 as an input. The input prompt 222 may be provided via an input device (e.g., a keyboard) via the I/O module of the server, such as I/O module 146. The prompt 222 may be previously unknown to the SFT ML model 215, e.g., the labelers may generate new prompt data, the prompt 222 may include testing data stored on database 126, and/or any other suitable prompt data. The SFT ML model 215 may generate multiple, different output responses 224A, 224B, 224C, 224D to the single prompt 222. The server 204 may output the responses 224A, 224B, 224C, 224D via an I/O module (e.g., I/O module 146) to a user interface device, such as a display (e.g., as text responses), a speaker (e.g., as audio/voice responses), and/or any other suitable manner of output of the responses 224A, 224B, 224C, 224D for review by the data labelers.

The data labelers may provide feedback via the server 204 on the responses 224A, 224B, 224C, 224D when ranking 226 them from best to worst based upon the prompt-response pairs. The data labelers may rank 226 the responses 224A, 224B, 224C, 224D by labeling the associated data. The ranked prompt-response pairs 228 may be used to train the reward model 220. In one aspect, the server 204 may load the reward model 220 via the ML module (e.g., the ML module 140) and train the reward model 220 using the ranked response pairs 228 as input. The reward model 220 may provide as an output the scalar reward 225.

In one aspect, the scalar reward **225** may include a value numerically representing a human preference for the best and/or most expected response to a prompt, i.e., a higher scalar reward value may indicate the service provider and/or insurance agent is more likely to prefer that response, and a lower scalar reward may indicate that the service provider and/or insurance agent is less likely to prefer that response. For example, inputting the “winning” prompt-response (i.e., input-output) pair data to the reward model **220** may generate a winning reward. Inputting a “losing” prompt-response pair data to the same reward model **220** may generate a losing reward. The reward model **220** and/or scalar reward **236** may be updated based upon labelers ranking **226** additional prompt-response pairs generated in response to additional prompts **222**.

In one example, a data labeler may provide to the SFT ML model **215** as an input prompt **222**, “Describe the sky.” The input may be provided by the labeler via the user device **102** over network **110** to the server **204** running a chatbot application utilizing the SFT ML model **215**. The SFT ML model **215** may provide as output responses to the labeler via the user device **102**: (i) “the sky is above” **224A**; (ii) “the sky includes the atmosphere and may be considered a place between the ground and outer space” **224B**; and (iii) “the sky is heavenly” **224C**. The data labeler may rank **226**, via labeling the prompt-response pairs, prompt-response pair **222/224B** as the most preferred answer; prompt-response pair **222/224A** as a less preferred answer; and prompt-response **222/224C** as the least preferred answer. The labeler may rank **226** the prompt-response pair data in any suitable manner. The ranked prompt-response pairs **228** may be provided to the reward model **220** to generate the scalar reward **225**.

While the reward model **220** may provide the scalar reward **225** as an output, the reward model **220** may not generate a response (e.g., text). Rather, the scalar reward **225** may be used by a version of the SFT ML model **215** to generate more accurate responses to prompts, i.e., the SFT model **215** may generate the response such as text to the prompt, and the reward model **220** may receive the response to generate a scalar reward **225** of how well humans perceive it. Reinforcement learning may optimize the SFT model **215** with respect to the reward model **220** which may realize the configured ML chatbot model **250**.

RLHF to Train the ML Chatbot Model

In one aspect, the server **206** may train the ML chatbot model **250** (e.g., via the ML module **140**) to generate a response **234** to a random, new and/or previously unknown service provider and/or insurance agent prompt **232**. To generate the response **234**, the ML chatbot model **250** may use a policy **235** (e.g., algorithm) which it learns during training of the reward model **220**, and in doing so may advance from the SFT model **215** to the ML chatbot model **250**. The policy **235** may represent a strategy that the ML chatbot model **250** learns to maximize its reward **225**. As discussed herein, based upon prompt-response pairs, a human labeler may continuously provide feedback to assist in determining how well the ML chatbot’s **250** responses match expected responses to determine rewards **225**. The rewards **225** may feed back into the ML chatbot model **250** to evolve the policy **235**. Thus, the policy **235** may adjust the parameters of the ML chatbot model **250** based upon the rewards **225** it receives for generating good responses. The policy **235** may update as the ML chatbot model **250** provides responses **234** to additional prompts **232**.

In one aspect, the response **234** of the ML chatbot model **250** using the policy **235** based upon the reward **225** may be

compared **238** to the SFT ML model **215** (which may not use a policy) response **236** of the same prompt **232**. The server **206** may compute a penalty **240** based upon the comparison **238** of the responses **234**, **236**. The penalty **240** may reduce the distance between the responses **234**, **236**, i.e., a statistical distance measuring how one probability distribution is different from a second, in one aspect the response **234** of the ML chatbot model **250** versus the response **236** of the SFT model **215**. Using the penalty **240** to reduce the distance between the responses **234**, **236** may avoid a server over-optimizing the reward model **220** and deviating too drastically from the human-intended/preferred response. Without the penalty **240**, the ML chatbot model **250** optimizations may result in generating responses **234** which are unreasonable but may still result in the reward model **220** outputting a high reward **225**.

In one aspect, the responses **234** of the ML chatbot model **250** using the current policy **235** may be passed by the server **206** to the rewards model **220**, which may return the scalar reward **225**. The ML chatbot model **250** response **234** may be compared **238** to the SFT ML model **215** response **236** by the server **206** to compute the penalty **240**. The server **206** may generate a final reward **242** which may include the scalar reward **225** offset and/or restricted by the penalty **240**. The final reward **242** may be provided by the server **206** to the ML chatbot model **250** and may update the policy **235**, which in turn may improve the functionality of the ML chatbot model **250**.

To optimize the ML chatbot **250** over time, RLHF via the human labeler feedback may continue ranking **226** responses of the ML chatbot model **250** versus outputs of earlier/other versions of the SFT ML model **215**, i.e., providing positive or negative rewards **225**. The RLHF may allow the servers (e.g., servers **204**, **206**) to continue iteratively updating the reward model **220** and/or the policy **235**. As a result, the ML chatbot model **250** may be retrained and/or fine-tuned based upon the human feedback via the RLHF process, and throughout continuing conversations may become increasingly efficient.

Although multiple servers **202**, **204**, **206** are depicted in the exemplary block and logic diagram **200**, each providing one of the three steps of the overall ML chatbot model **250** training, fewer and/or additional servers may be utilized and/or may provide the one or more steps of the ML chatbot model **250** training. In one aspect, one server may provide the entire ML chatbot model **250** training.

Exemplary ML Model for Sump Pump Fault Detection Analysis

In one embodiment, determining whether a sump pump is faulty may use ML.

FIG. 2B schematically illustrates how an ML model may generate a sump pump fault detection analysis. Some of the blocks in FIG. 2B represent hardware and/or software components (e.g., block **265**), other blocks represent data structures or memory storing these data structures, registers, or state variables (e.g., blocks **260**), and other blocks represent output data (e.g., block **280**). Input and output signals are represented by arrows.

An ML engine **265** may include one or more hardware and/or software components, such as the MLTM **142** and/or the MLOM **144**, to obtain, create, (re)train, operate and/or save one or more ML models **290**. To generate an ML model **290**, the ML engine **265** may use training data **260**.

As described herein, the server such as server **105** may obtain and/or have available various types of training data **260** (e.g., stored on database **126** of server **105**). In an aspect, the training data **260** may be labeled to aid in training, retrain-

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ing and/or fine-tuning the ML model **290**. The training data **260** may include historical operating and failure data for one or more sump pumps. The historical operating and failure data may comprise model number, age, hours of service, current drawn, operating temperature, flow rate, and/or any other suitable information about the sump pumps at or near the time of failure. The training data **260** may be in a structured or unstructured format. New training data **260** may be used to retrain or update the ML model **290**.

While the example training data includes indications of various types of training data **260**, this is merely an example for ease of illustration only. The training data **260** may include any suitable data that may indicate associations between sump pump operating data and detection of a fault.

In an aspect, the server may continuously update the training data **260**, e.g., based upon obtaining data sources related to the data collected from prior sump pump failures, or any other training data. Subsequently, the ML model **290** may be retrained/fine-tuned based upon the updated training data **260**. Accordingly, the generation of fault detection analyses may improve over time.

In an aspect, the ML engine **265** may process and/or analyze the training data **260** (e.g., via MLTM **142**) to train the ML model **290** to generate the fault detection analysis **280**. The ML model **290** may be trained to generate the fault detection analysis **280** via a neural network, deep learning model, Transformer-based model, generative pretrained transformer (GPT), generative adversarial network (GAN), regression model, k-nearest neighbor algorithm, support vector regression algorithm, and/or random forest algorithm, although any type of applicable ML model/algorithm may be used, including training using one or more of supervised learning, unsupervised learning, semi-supervised learning, and/or reinforcement learning.

Once trained, the ML model **290** may perform operations on one or more data inputs to produce a desired data output. In one aspect, the ML model **290** may be loaded at runtime (e.g., by the MLOM **144**) from a database (e.g., the database **126** of the server **105**) to process sump pump operating data **270** and/or precipitation forecast **275** input. The server, such as server **105**, may obtain the process sump pump operating data **270** and/or precipitation forecast **275** and use them as input to generate the fault detection analysis **280**. The server may use the precipitation forecast **275** to predict whether the sump pump will be able to keep up with the forecasted precipitation. In one aspect, the server may obtain the sump pump operating data **270** via the client device **102**, a website, the chatbot **150**, or any other suitable user device. In one aspect, the server may obtain the precipitation forecast **275** via the client device **102**, a website, the chatbot **150**, or any other suitable source.

In one aspect, the sump pump operating data **270** may comprise model number, age, hours of service, current drawn, operating temperature, flow rate, and/or any other suitable information about a sump pump. In one aspect, the precipitation forecast **275** may comprise a probability of precipitation and/or an amount of precipitation over a number of upcoming days.

Once the fault detection analysis **280** is generated by ML model **290**, it may be provided to the client device **102**, server **105**, or to another user device. For example, the server **105** may provide the fault detection analysis **280** via a mobile app to mobile device, in an email, a website, via a chatbot (such as the chatbot **295**), and/or in any other suitable manner.

Exemplary Computer System for Sump Pump Monitoring and Service Provider Notification

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FIG. 3 depicts an exemplary environment **300** in which methods and systems for sump pump fault monitoring and service provider notification may be performed, in accordance with various aspects discussed herein.

In one aspect, a structure **310** may comprise one or more sump pumps **320**. In one aspect, the structure **310** may be a house, apartment, condominium, or any other type of residential dwelling. In another aspect, the structure **310** may be a high rise tower, shopping center, data center, factory, warehouse, or any other type of commercial building. The structure **310** may comprise subterranean levels, such as a basement, parking garage, loading dock, etc.

In one aspect, the sump pump **320** may be powered by electricity, steam, water pressure, or compressed air. The sump pump **320** may run continuously, manually, or automatically via a water level sensing switch. In one aspect, the sump pump **320** may comprise one or more operation sensors that detect if the sump pump **320** is faulty. The operation sensors may be integrated into the sump pump **320** or may comprise separate components. The operation sensors may detect if the sump pump **320** is inoperable or impaired. The operation sensors may measure an electrical current drawn by the sump pump **320**, measure an operating temperature of the sump pump **320**, and/or measure water pressure or a flow rate at an output of the sump pump **320**. The operation sensors may detect a water level in a sump.

In one aspect, exemplary environment **300** may comprise one or more monitoring devices **102**. The monitoring device **102** may comprise an application for monitoring sump pump status and/or operating data. The monitoring device **102** may be located within the structure **310** or may be located remotely. The monitoring device **102** may receive analog and/or digital signals from the operation sensors via a wired connection. The monitoring device **102** may receive serial data from the operation sensors via a protocol such as I2C, SPI, RS232, or USB. The monitoring device **102** may receive and/or send network communications with the operation sensors via a wired protocol, such as Ethernet. The monitoring device **102** may receive and/or send network communications with the operation sensors via a wireless protocol, such as cellular, WiFi, Bluetooth, Zigbee, or LoRaWAN. The monitoring device **102** may receive status and/or operating data from the operation sensors. The monitoring device **102** may determine that the sump pump **320** is faulty based upon the received status and/or an alarm from a water sensor. In one aspect, the monitoring device **102** may transmit a sump pump fault alert to the server **105**. The monitoring device **102** may transmit the sump pump fault alert automatically (without human intervention) or after receiving a confirmation from a user.

In one aspect, the monitoring device **102** may receive weather data. For example, the monitoring device **102** may receive weather data from a weather server **360**. The weather server **360** may be operated by a government entity, news organization, security company, or any other organization. The weather server **360** may transmit notifications of weather data to the monitoring device **102**. The monitoring device **102** may periodically poll the weather server **360** and request weather data. The weather data may comprise a weather forecast and/or weather alerts, such as a flood alert. The weather data may indicate an occurrence of precipitation and/or precipitation exceeding a specified amount.

In one aspect, the monitoring device **102** may periodically transmit sump pump operating data and/or weather data to the server **105**. The monitoring device **102** may transmit a location of the structure **310**, a description of the structure

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310, a description of the sump pump 320, the sump pump operating data, and/or other relevant information.

In one aspect, exemplary environment 300 may comprise one or more servers 105. The server 105 may comprise a chatbot 150 and/or an ML model 290. The server 105 may determine that the sump pump 320 is faulty based upon the operating data received from the monitoring device. The server 105 may determine that the sump pump 320 is faulty based upon the received operating data and a weather forecast predicting precipitation.

The server 105 may comprise or retrieve information from a database of sump pump service providers. The server 105 may also retrieve a list of sump pump service providers from an online source, such as a search engine or a directory. The server may comprise or retrieve information from a database of information about structures, including the structure 310, and/or information about sump pumps, such as the sump pump 320. After receiving the sump pump fault alert from the monitoring device 102 or after determining the sump pump 320 is faulty, the server 105 may identify a subset of sump pump service providers 340A-340N within a certain geographic area associated with the structure 310.

The server 105 may generate one or more requests for information via a chatbot 150. In one aspect, the chatbot 150 is an ML chatbot, although the chatbot 150 may be an AI chatbot, a voice bot and/or any other suitable chatbot/voice bot as described herein. The server 105 may select an appropriate chatbot based upon the method of communication with the service providers.

In one aspect, the chatbot 150 may initiate communications with one or more of the of sump pump service providers 340A-340N to request sump pump repair and/or replacement. The chatbot 150 may communicate with one service provider at a time or with a plurality of service providers simultaneously. The chatbot 150 may communicate with the sump pump service providers 340A-340N via (i) audio (e.g., a telephone call), (ii) text messages (e.g., short messaging/SMS, multimedia messaging/MMS, iPhone iMessages, etc.), (iii) instant messages (e.g., real-time messaging such as a chat window), (iv) video such as video conferencing, and/or any other suitable communication means. The chatbot 150 may communicate with a human and/or another chatbot. The chatbot 150 may operate in a conversational manner and provide and collect information without any human intervention.

In one aspect, the chatbot 150 may receive utterances via an audio connection with one or more of the sump pump service providers 340A-340N (e.g., as part of a voice call initiated by the chatbot 150). The chatbot 150 may transcribe the audio utterances into unformatted text. The NLP module 148 may convert the unformatted text into structured input data. The server 150 may store the structured input data in the database 126. The ML module 140 may generate structured output data based on the input data. The NLP module 148 may convert the structured output data into unformatted text. The chatbot may convert the unformatted text into audio data and output the audio data, e.g., a follow up question, to the service provider.

The chatbot 150 may provide the location of the structure 310, the description of the structure 310, the description of the sump pump 320, the received operating data, and/or other relevant information to the service providers 340A-340N. The chatbot 150 may ask the service providers 340A-340N questions, such as what sump pump models they have in stock, what sump pump models they service, schedule availability, estimated price, and/or other questions to gather relevant information. For example, the chatbot 150

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may ask the service provider 340A to confirm it repairs and/or replaces sump pumps. The chatbot 150 may provide information about the sump pump 320, such as the manufacturer and/or model number. The chatbot 150 may provide information about the structure 310, such as whether it is a residential house or a commercial building, and the address of the structure 310. The chatbot 150 may ask the service provider 340A what dates(s) and time(s) are available for the service provider 340A to send a technician to the structure 310 to repair or replace the sump pump 320. The chatbot 150 may ask the service provider 340A for an estimated charge for repairing and/or replacing the sump pump 320, including parts and labor. The chatbot 150 may negotiate a lower price with the service providers 340A-340N based upon the service provider 340A's estimated charge. For example, the chatbot 150 may ask service provider 340B to match an estimated charge provided by service provider 340A.

The server 105 may collect information from the service providers 340A-340N. The server 105 may analyze and/or process the collected information to interpret, understand and/or extract relevant information within one or more responses from the service provider. In one aspect, the server 105 may select one of the service providers 340A-340N based upon the relevant information. For example, the server 105 may select the service provider having the lowest price, the earliest schedule availability, or a combination of factors from the relevant information. The chatbot 150 may initiate communication with the selected service provider to order sump pump repair and/or replacement. In another aspect, the server 105 may select a subset of one or more of the service providers 340A-340N based upon the relevant information. For example, the server 105 may select the subset of one or more service providers having the lowest prices, the earliest schedule availabilities, or a combination of factors from the relevant information. The chatbot 150 may initiate communication with a user device 330 and provide the selected subset of service providers and the relevant information from that selected subset of service providers.

The user device 330 may comprise one or more of desktop computers, laptops, smartphones, wearables, smart watches, smart contact lenses, smart glasses, augmented reality glasses, virtual reality headsets, mixed or extended reality glasses or headsets, and/or any other suitable communication device. The chatbot 150 may communicate with the user device 330 via audio, text messages, instant messages, video, e-mail, application notifications, and/or any other suitable communication means. A user may select one of the service providers via the user device 330, which may communicate the service provider selection to the chatbot 150. The chatbot 150 may initiate communication with the selected service provider to order sump pump repair and/or replacement. The selected service provider and the relevant information from selected service provider may be stored by the server 105 in memory, such as the memory 122 and/or database 126.

In one aspect, the server 105, e.g., via NLP model 148 and/or the chatbot 150, may analyze the communication sessions with the service providers 340A-340N and/or the user for indications of sentiment, such as the emotion of the service provider or user (e.g., upset, stressed, calm, frustrated, impatient, etc.). Other types of suitable analysis and/or analytics may be obtained from the communication session information.

In one aspect, types of data the communication sessions may generate may include the length of the session, which may indicate how effective the chatbot 150 is at gathering or providing necessary information (e.g., a short session may

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not gather enough information; a long session may provide too much and/or inaccurate information). Another type of data the session may generate may include how many requests were generated by the chatbot 150, which may also indicate the quality and/or effectiveness of the session (e.g., too few questions may not gather enough information and too many questions may indicate ineffectiveness of the questions being asked). The number of requests may also indicate when the session warrants termination, for example the chatbot 150 may no longer have any requests to generate which may indicate all information relevant may be gathered. Any suitable analytics and/or data may be generated and/or analyzed from the session which may indicate the quality and/or effectiveness of the session and/or chatbot 150.

In one aspect, the chatbot 150 may determine a confidence level at one or more instances during the communication session. The confidence level and/or score, which may be a number between 0 and 1, may represent the likelihood that the output of the chatbot 150 is correct and will satisfy the service provider's or the user's request. As the output of the chatbot 150 may include one or more predictions, each prediction may have a confidence score wherein the higher the score, the more confident the chatbot 150 is that the prediction may satisfy the service provider's or the user's request. In conversational AI/ML which may include the chatbot 150, one or more stages may process the request and/or input of the service provider or the user. In one aspect, during NLU, the chatbot 150 may predict the service provider or user intent (what the service provider or user is looking for) from an utterance/prompt (what the service provider or user may say or type). In one aspect, during sentiment and/or emotion analysis, the chatbot 102 may predict the sentiment (e.g., positive, negative, or neutral) and/or the emotion of the service provider or user based upon the service provider or user utterance and/or prompts (back and forth between the service provider or user and the chatbot 150) transcript. In one aspect, during NLG, the chatbot 150 may predict what to respond based upon the service provider or user utterance/prompt. One or more of these predictions may have an associated confidence score/level.

In one aspect, the server 105 and/or chatbot 150 may determine the confidence level based upon the interactions between the chatbot 150 and the service provider or user during the communication session, e.g., how accurately does it seem the chatbot 150 is able to interpret the service provider or user responses, how effective are chatbot 150 requests, and/or other suitable metrics and/or analysis of the communication session to determine the confidence level of the chatbot 150. In one aspect, the chatbot 150 confidence level may be compared to a threshold confidence level (e.g., which may also be a value between 0 and 1) by the server 105 and/or chatbot 150. If the chatbot 150 confidence level falls below the threshold, one or more actions may be taken by the server 105 and/or chatbot 150, such as ending the communication session, using a different chatbot 150 to continue the communication session (e.g., one which may be trained to more effectively assist the service provider or user), and/or any other suitable action as may be described herein.

Exemplary Computer System for Flood Monitoring and Service Provider Notification

FIG. 4 depicts an exemplary environment 400 in which methods and systems for flood monitoring and service provider notification may be performed, in accordance with various aspects discussed herein.

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In one aspect, a structure 410 may comprise one or more sensors 450. In one aspect, the structure 410 may be a house, apartment, condominium, or any other type of residential dwelling. In another aspect, the structure 410 may be a high rise tower, shopping center, data center, factory, warehouse, or any other type of commercial building. The structure 410 may comprise subterranean levels, such as a basement, parking garage, loading dock, etc.

In one aspect, the sensors 450 may be water sensors, such as a conductive sensor, a capacitive sensor, an optical sensor, or a float switch. The sensors 450 may detect an interruption of electrical power to the structure.

In one aspect, exemplary environment 400 may comprise one or more monitoring devices 102. The monitoring device 102 may comprise an application for monitoring data from the sensors 350. The monitoring device 102 may be located within the structure 410 or may be located remotely. The monitoring device 102 may receive analog and/or digital signals from the sensors 450 via a wired connection. The monitoring device 102 may receive serial data from the operation sensors via a protocol such as I2C, SPI, RS232, or USB. The monitoring device 102 may receive and/or send network communications with the sensors 450 via a wired protocol, such as Ethernet. The monitoring device 102 may receive and/or send network communications with the sensors 450 via a wireless protocol, such as cellular, WiFi, Bluetooth, Zigbee, or LoRaWAN. The monitoring device 102 may receive water detection alarms and/or electrical power interruption alarms from the sensors 450. The monitoring device 102 may determine that the structure 410 is experiencing a flood based upon the received water detection alarms.

In one aspect, the monitoring device 102 may determine that the structure 410 is experiencing a flood based upon received weather data. The monitoring device 102 may determine that the structure 410 is experiencing a flood based upon received weather data and electrical power interruption alarms from the sensors 450. For example, the monitoring device 102 may receive weather data from a weather server 460. The weather server 460 may be operated by a government entity, news organization, security company, or any other organization. The weather server 460 may transmit notifications of weather data to the monitoring device 102. The monitoring device 102 may periodically poll the weather server 460 and request weather data. The weather data may comprise a weather forecast and/or weather alerts, such as a flood alert. The weather data may indicate an occurrence of precipitation and/or precipitation exceeding a specified amount.

In one aspect, the monitoring device 102 may transmit a flood alert to the server 105. The monitoring device 102 may transmit the flood alert automatically, i.e., without human intervention, or after receiving a confirmation from a user. The monitoring device 102 may transmit a location of the structure 410, a description of the structure 410, alarm information from the sensors 450, and/or other relevant information.

In one aspect, exemplary environment 400 may comprise one or more servers 105. The server 105 may comprise a chatbot 150. The server 105 may comprise or retrieve information from a database of flood remediation service providers. The server 105 may also retrieve a list of flood remediation service providers from an online source, such as a search engine or a directory. The server may comprise or retrieve information from a database of information about structures, including the structure 410. After receiving the flood alert from the monitoring device 102, the server 105

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may identify a subset of flood remediation service providers **440A-440N** within a certain geographic area associated with the structure **410**.

The server **105** may generate one or more requests for information via a chatbot **150**. In one aspect, the chatbot **150** is an ML chatbot, although the chatbot **150** may be an AI chatbot, a voice bot and/or any other suitable chatbot/voice bot as described herein. The server **105** may select an appropriate chatbot based upon the method of communication with the service providers.

In one aspect, the chatbot **150** may initiate communications with one or more of the of flood remediation service providers **440A-440N** to request flood remediation. The chatbot **150** may communicate with one service provider at a time or with a plurality of service providers simultaneously. The chatbot **150** may communicate with the flood remediation providers **440A-440N** via (i) audio (e.g., a telephone call), (ii) text messages (e.g., short messaging/SMS, multimedia messaging/MMS, iPhone iMessages, etc.), (iii) instant messages (e.g., real-time messaging such as a chat window), (iv) video such as video conferencing, and/or any other suitable communication means. The chatbot **150** may communicate with a human and/or another chatbot. The chatbot **150** may operate in a conversational manner and provide and collect information without any human intervention.

In one aspect, the chatbot **150** may receive utterances via an audio connection with one or more of the sump pump service providers **440A-440N** (e.g., as part of a voice call initiated by the chatbot **150**). The chatbot **150** may transcribe the audio utterances into unformatted text. The NLP module **148** may convert the unformatted text into structured input data. The server **105** may store the structured input data in the database **126**. The ML module **140** may generate structured output data based on the input data. The NLP module **148** may convert the structured output data into unformatted text. The chatbot may convert the unformatted text into audio data and output the audio data, e.g., a follow up question, to the service provider.

The chatbot **150** may provide the location of the structure **410**, the description of the structure **410**, the received flood alert information, and/or other relevant information to the service providers **440A-440N**. The chatbot **150** may ask the service providers **440A-440N** questions, such as whether they have water pumps available, schedule availability, estimated price, and/or other questions to gather relevant information. For example, the chatbot **150** may ask the service provider **440A** to confirm it provides flood remediation services. The chatbot **150** may provide information gathered from the sensors, such as which portions of the structure **410** are flooded. The chatbot **150** may provide information about the structure **310**, such as whether it is a residential house or a commercial building, and the address of the structure **310**. The chatbot **150** may ask the service provider **440A** what dates(s) and time(s) are available for the service provider **440A** to send a technician to the structure **410** to begin remediation work. The chatbot **150** may ask the service provider **440A** for an estimated charge for the flood remediation. The chatbot **150** may negotiate a lower price with the service providers **440A-440N** based on the service provider **440A**'s estimated charge. For example, the chatbot **150** may ask service provider **440B** to match an estimated charge provided by service provider **440A**.

The server **105** may collect information from the service providers **440A-440N**. The server **105** may analyze and/or process the collected information to interpret, understand and/or extract relevant information within one or more

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responses from the service provider. In one aspect, the server **105** may select one of the service providers **440A-440N** based upon the relevant information. For example, the server **105** may select the service provider having the lowest price, the earliest schedule availability, or a combination of factors from the relevant information. The chatbot **150** may initiate communication with the selected service provider to order flood remediation services. In another aspect, the server **105** may select a subset of one or more of the service providers **440A-440N** based upon the relevant information. For example, the server **105** may select the subset of one or more service providers having the lowest prices, the earliest schedule availabilities, or a combination of factors from the relevant information. The chatbot **150** may initiate communication with a user device **430** and provide the selected subset of service providers and the relevant information from that selected subset of service providers.

The user device **430** may comprise one or more of desktop computers, laptops, smartphones, wearables, smart watches, smart contact lenses, smart glasses, augmented reality glasses, virtual reality headsets, mixed or extended reality glasses or headsets, and/or any other suitable communication device. The chatbot **150** may communicate with the user device **430** via audio, text messages, instant messages, video, e-mail, application notifications, and/or any other suitable communication means. A user may select one of the service providers via the user device **430**, which may communicate the service provider selection to the chatbot **150**. The chatbot **150** may initiate communication with the selected service provider to order flood remediation services. The selected service provider and the relevant information from selected service provider may be stored by the server **105** in memory, such as the memory **122** and/or database **126**.

In one aspect, the server **105**, e.g., via NLP model **148** and/or the chatbot **150**, may analyze the communication sessions with the service providers **440A-440N** and/or the user for indications of sentiment, such as the emotion of the service provider or user (e.g., upset, stressed, calm, frustrated, impatient, etc.). Other types of suitable analysis and/or analytics may be obtained from the communication session information.

In one aspect, types of data the communication sessions may generate may include the length of the session, which may indicate how effective the chatbot **150** is at gathering or providing necessary information (e.g., a short session may not gather enough information; a long session may provide too much and/or inaccurate information). Another type of data the session may generate may include how many requests were generated by the chatbot **150**, which may also indicate the quality and/or effectiveness of the session (e.g., too few questions may not gather enough information and too many questions may indicate ineffectiveness of the questions being asked). The number of requests may also indicate when the session warrants termination, for example the chatbot **150** may no longer have any requests to generate which may indicate all information relevant may be gathered. Any suitable analytics and/or data may be generated and or analyzed from the session which may indicate the quality and/or effectiveness of the session and/or chatbot **150**.

In one aspect, the chatbot **150** may determine a confidence level at one or more instances during the communication session. The confidence level and/or score, which may be a number between 0 and 1, may represent the likelihood that the output of the chatbot **150** is correct and will satisfy the service provider's or the user's request. As the output of the

chatbot **150** may include one or more predictions, each prediction may have a confidence score wherein the higher the score, the more confident the chatbot **150** is that the prediction may satisfy the service provider's or the user's request. In conversational AI/ML which may include the chatbot **150**, one or more stages may process the request and/or input of the service provider or the user. In one aspect, during NLU, the chatbot **150** may predict the service provider or user intent (what the service provider or user is looking for) from an utterance/prompt (what the service provider or user may say or type). In one aspect, during sentiment and/or emotion analysis, the chatbot **102** may predict the sentiment (e.g., positive, negative, or neutral) and/or the emotion of the service provider or user based upon the service provider or user utterance and/or prompt (back and forth between the service provider or user and the chatbot **150**) transcript. In one aspect, during NLG, the chatbot **150** may predict what to respond based upon the service provider or user utterance/prompt. One or more of these predictions may have an associated confidence score/level.

In one aspect, the server **105** and/or chatbot **150** may determine the confidence level based upon the interactions between the chatbot **150** and the service provider or user during the communication session, e.g., how accurately does it seem the chatbot **150** is able to interpret the service provider or user responses, how effective are chatbot **150** requests, and/or other suitable metrics and/or analysis of the communication session to determine the confidence level of the chatbot **150**. In one aspect, the chatbot **150** confidence level may be compared to a threshold confidence level (e.g., which may also be a value between 0 and 1) by the server **105** and/or chatbot **150**. If the chatbot **150** confidence level falls below the threshold, one or more actions may be taken by the server **105** and/or chatbot **150**, such as ending the communication session, using a different chatbot **150** to continue the communication session (e.g., one which may be trained to more effectively assist the service provider or user), and/or any other suitable action as may be described herein.

Exemplary Computer System for Flood Monitoring and Insurance Provider Notification

FIG. 5 depicts an exemplary environment **500** in which methods and systems for flood monitoring and insurance provider notification may be performed, in accordance with various aspects discussed herein.

In one aspect, a structure **510** may comprise one or more sensors **550**. In one aspect, the structure **510** may be a house, apartment, condominium, or any other type of residential dwelling. In another aspect, the structure **510** may be a high rise tower, shopping center, data center, factory, warehouse, or any other type of commercial building. The structure **510** may comprise subterranean levels, such as a basement, parking garage, loading dock, etc.

In one aspect, the sensors **550** may be water sensors, such as a conductive sensor, a capacitive sensor, an optical sensor, or a float switch. The sensors **550** may detect an interruption of electrical power to the structure.

In one aspect, exemplary environment **500** may comprise one or more monitoring devices **102**. The monitoring device **102** may comprise an application for monitoring data from the sensors **550**. The monitoring device **102** may be located within the structure **510** or may be located remotely. The monitoring device **102** may receive analog and/or digital signals from the sensors **550** via a wired connection. The monitoring device **102** may receive serial data from the operation sensors via a protocol such as I2C, SPI, RS232, or

USB. The monitoring device **102** may receive and/or send network communications with the sensors **550** via a wired protocol, such as Ethernet. The monitoring device **102** may receive and/or send network communications with the sensors **550** via a wireless protocol, such as cellular, WiFi, Bluetooth, Zigbee, or LoRaWAN. The monitoring device **102** may receive water detection alarms and/or electrical power interruption alarms from the sensors **550**. The monitoring device **102** may determine that the structure **510** is experiencing a flood based upon the received water detection alarms.

In one aspect, the monitoring device **102** may determine that that the structure **510** is experiencing a flood based upon received weather data. The monitoring device **102** may determine that that the structure **510** is experiencing a flood based upon received weather data and electrical power interruption alarms from the sensors **550**. For example, the monitoring device **102** may receive weather data from a weather server **560**. The weather server **560** may be operated by a government entity, news organization, security company, or any other organization. The weather server **560** may transmit notifications of weather data to the monitoring device **102**. The monitoring device **102** may periodically poll the weather server **560** and request weather data. The weather data may comprise a weather forecast and/or weather alerts, such as a flood alert. The weather data may indicate an occurrence of precipitation and/or precipitation exceeding a specified amount.

In one aspect, the monitoring device **102** may transmit a flood alert to the server **105**. The monitoring device **102** may transmit the flood alert automatically, i.e., without human intervention, or after receiving a confirmation from a user. The monitoring device **102** may transmit a location of the structure **510**, a description of the structure **510**, alarm information from the sensors **550**, and/or other relevant information.

In one aspect, exemplary environment **500** may comprise one or more servers **105**. The server **105** may comprise a chatbot **150**. The server **105** may comprise or retrieve insurance information from a database regarding the structure **510**, including insurance provider contact information and flood coverage. The server may comprise or retrieve information from a database of information about structures, including the structure **510**. After receiving the flood alert from the monitoring device **102**, the server **105** may identify the insurance provider **540** with the structure **510**.

The server **105** may generate one or more requests for information via a chatbot **150**. In one aspect, the chatbot **150** is an ML chatbot, although the chatbot **150** may be an AI chatbot, a voice bot and/or any other suitable chatbot/voice bot as described herein. The server **105** may select an appropriate chatbot based upon the method of communication with the service providers.

In one aspect, the chatbot **150** may initiate communications with the insurance provider **540** to initiate a flood reimbursement claim. The insurance provider **540** may be an insurance company that issued the policy on the structure **510** or an independent insurance agent. The chatbot **150** may communicate with the insurance provider **540** via (i) audio (e.g., a telephone call), (ii) text messages (e.g., short messaging/SMS, multimedia messaging/MMS, iPhone iMessages, etc.), (iii) instant messages (e.g., real-time messaging such as a chat window), (iv) video such as video conferencing, and/or any other suitable communication means. The chatbot **150** may communicate with a human and/or

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another chatbot. The chatbot **150** may operate in a conversational manner and provide and collect information without any human intervention.

In one aspect, the chatbot **150** may receive utterances via an audio connection from the insurance provider **540** (e.g., as part of a voice call initiated by the chatbot **150**). The chatbot **150** may transcribe the audio utterances into unformatted text. The NLP module **148** may convert the unformatted text into structured input data. The server **150** may store the structured input data in the database **126**. The ML module **140** may generate structured output data based on the input data. The NLP module **148** may convert the structured output data into unformatted text. The chatbot may convert the unformatted text into audio data and output the audio data, e.g., a follow up question, to the insurance provider **540**.

The chatbot **150** may provide the location of the structure **510**, the description of the structure **510**, insurance policy number associated with the structure **510**, the received flood alert information, and/or other relevant information to the insurance provider **540**. The chatbot **150** may ask the insurance provider **540** questions, such as when an adjuster will be available to inspect the structure **510**, will the insurance provider **540** provide a flood remediation contractor, and/or other questions to gather relevant information. The server **105** may collect information from the insurance provider **540**. The server **105** may analyze and/or process the collected information to interpret, understand and/or extract relevant information within one or more responses from the insurance provider **540**.

The chatbot may provide the relevant information to a user device **530**. The chatbot **150** may communicate with the user device **530** via audio, text messages, instant messages, video, e-mail, application notifications, and/or any other suitable communication means. The user device **530** may comprise one or more of desktop computers, laptops, smartphones, wearables, smart watches, smart contact lenses, smart glasses, augmented reality glasses, virtual reality headsets, mixed or extended reality glasses or headsets, and/or any other suitable communication device.

In one aspect, the server **105**, e.g., via NLP model **148** and/or the chatbot **150**, may analyze the communication session with the insurance provider **540** and/or the user for indications of sentiment, such as the emotion of the insurance provider or user (e.g., upset, stressed, calm, frustrated, impatient, etc. Other types of suitable analysis and/or analytics may be obtained from the communication session information.

In one aspect, types of data the communication sessions may generate may include the length of the session, which may indicate how effective the chatbot **150** is at gathering or providing necessary information (e.g., a short session may not gather enough information; a long session may provide too much and/or inaccurate information). Another type of data the session may generate may include how many requests were generated by the chatbot **150**, which may also indicate the quality and/or effectiveness of the session (e.g., too few questions may not gather enough information and too many questions may indicate ineffectiveness of the questions being asked). The number of requests may also indicate when the session warrants termination, for example the chatbot **150** may no longer have any requests to generate which may indicate all information relevant may be gathered. Any suitable analytics and/or data may be generated and or analyzed from the session which may indicate the quality and/or effectiveness of the session and/or chatbot **150**.

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In one aspect, the chatbot **150** may determine a confidence level at one or more instances during the communication session. The confidence level and/or score, which may be a number between 0 and 1, may represent the likelihood that the output of the chatbot **150** is correct and will satisfy the insurance provider **340**'s or the user's request. As the output of the chatbot **150** may include one or more predictions, each prediction may have a confidence score wherein the higher the score, the more confident the chatbot **150** is that the prediction may satisfy the insurance provider **340**'s or the user's request. In conversational AI/ML which may include the chatbot **150**, one or more stages may process the request and/or input of the insurance provider **340** or the user. In one aspect, during NLU, the chatbot **150** may predict the insurance provider **340** or user intent (what the service provider or user is looking for) from an utterance/prompt (what the service provider or user may say or type). In one aspect, during sentiment and/or emotion analysis, the chatbot **102** may predict the sentiment (e.g., positive, negative, or neutral) and/or the emotion of the insurance provider **540** or user based upon the insurance provider **540** or user utterance and/or prompt (back and forth between the service provider or user and the chatbot **150**) transcript. In one aspect, during NLG, the chatbot **150** may predict what to respond based upon the insurance provider **540** or user utterance/prompt. One or more of these predictions may have an associated confidence score/level.

In one aspect, the server **105** and/or chatbot **150** may determine the confidence level based upon the interactions between the chatbot **150** and the insurance provider **540** or user during the communication session, e.g., how accurately does it seem the chatbot **150** is able to interpret the insurance provider **540** or user responses, how effective are chatbot **150** requests, and/or other suitable metrics and/or analysis of the communication session to determine the confidence level of the chatbot **150**. In one aspect, the chatbot **150** confidence level may be compared to a threshold confidence level (e.g., which may also be a value between 0 and 1) by the server **105** and/or chatbot **150**. If the chatbot **150** confidence level falls below the threshold, one or more actions may be taken by the server **105** and/or chatbot **150**, such as ending the communication session, using a different chatbot **150** to continue the communication session (e.g., one which may be trained to more effectively assist the insurance provider **540** provider or user), and/or any other suitable action as may be described herein.

Exemplary Computer-Implemented Method for Flood Detection and Insurance Claim Filing

FIG. 6 depicts a flow diagram of an exemplary computer-implemented method **600** for detecting a flood and filing a claim with an insurance provider. One or more steps of the method **600** may be implemented as a set of instructions stored on a computer-readable memory and executable on one or more processors. The method **600** of FIG. 6 may be implemented via one or more systems, such as monitoring device **102** and/or server **105**.

In one embodiment, the computer-implemented method **600** may include training an ML chatbot (such as the ML chatbots **150** or **250**) with a training dataset and/or validating the ML chatbot with a validation dataset. The training dataset and/or the validation dataset may comprise historical data from past notices of flood events, transcripts of telephone and/or chat sessions with insurance providers, and/or user ratings of the chatbot output.

In one embodiment, the computer-implemented method **600** may include at block **610** detecting a flood event in a structure (such as structure **310**, **410**, or **510**). Detecting a

flood event may comprise receiving a signal from a water sensor (such as sensors **450** or **550**). Detecting a flood event may comprise receiving a notification of precipitation exceeding a specified amount. Detecting a flood event may comprise receiving a notification of an occurrence of precipitation and detecting an interruption of electricity to the structure.

In one embodiment, the computer-implemented method **600** at block **620** may include transmitting structure information and prompt for flood reimbursement to the ML chatbot (such as ML chatbot **150** or **250**). The structure information and prompt may be sent via a text message, application, e-mail, FTP, HTTP, HTTPS, and/or any other suitable communication method.

The prompt for flood reimbursement may cause the ML chatbot to file a flood reimbursement claim with an insurance provider having an insurance policy associated with the structure. The prompt for flood reimbursement may cause the ML chatbot to file a flood reimbursement claim with an insurance provider via telephone by converting a text output into a voice output. The prompt for flood reimbursement may cause the ML chatbot to file a flood reimbursement claim with an insurance provider by providing information associated with the structure.

In one embodiment, the computer-implemented method **600** may include causing the ML chatbot to request a predetermined payout amount from the insurance provider.

In one embodiment, the computer-implemented method **600** may include causing the ML chatbot to collect information from the insurance provider. The method may also include causing the ML chatbot to communicate the information to a user associated with the structure.

It should be understood that not all blocks of the exemplary flow diagram **600** are required to be performed. Moreover, the exemplary flow diagram **600** is not mutually exclusive (i.e., block(s) from exemplary flow diagram **600** may be performed in any particular implementation).

Additional Considerations

Although the text herein sets forth a detailed description of numerous different embodiments, it should be understood that the legal scope of the invention is defined by the words of the claims set forth at the end of this patent. The detailed description is to be construed as exemplary only and does not describe every possible embodiment, as describing every possible embodiment would be impractical, if not impossible. One could implement numerous alternate embodiments, using either current technology or technology developed after the filing date of this patent, which would still fall within the scope of the claims.

It should also be understood that, unless a term is expressly defined in this patent using the sentence “As used herein, the term ‘ ’ is hereby defined to mean . . . ” or a similar sentence, there is no intent to limit the meaning of that term, either expressly or by implication, beyond its plain or ordinary meaning, and such term should not be interpreted to be limited in scope based upon any statement made in any section of this patent (other than the language of the claims). To the extent that any term recited in the claims at the end of this disclosure is referred to in this disclosure in a manner consistent with a single meaning, that is done for sake of clarity only so as to not confuse the reader, and it is not intended that such claim term be limited, by implication or otherwise, to that single meaning. Finally, unless a claim element is defined by reciting the word “means” and a function without the recital of any structure, it is not intended that the scope of any claim element be interpreted based upon the application of 35 U.S.C. § 112(f).

Throughout this specification, plural instances may implement components, operations, or structures described as a single instance. Although individual operations of one or more methods may be illustrated and described as separate operations, one or more of the individual operations may be performed concurrently, and nothing requires that the operations be performed in the order illustrated. Structures and functionality presented as separate components in exemplary configurations may be implemented as a combined structure or component. Similarly, structures and functionality presented as a single component may be implemented as separate components. These and other variations, modifications, additions, and improvements fall within the scope of the subject matter herein.

Additionally, certain embodiments may be described herein as including logic or a number of routines, subroutines, applications, or instructions. These may constitute either software (code embodied on a non-transitory, tangible machine-readable medium) or hardware. In hardware, the routines, etc., are tangible units capable of performing certain operations and may be configured or arranged in a certain manner. In exemplary embodiments, one or more computer systems (e.g., a standalone, client or server computer system) or one or more hardware modules of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware module that operates to perform certain operations as described herein.

In various embodiments, a hardware module may be implemented mechanically or electronically. For example, a hardware module may comprise dedicated circuitry or logic that is permanently configured (e.g., as a special-purpose processor, such as a field programmable gate array (FPGA) or an application-specific integrated circuit (ASIC) to perform certain operations). A hardware module may also comprise programmable logic or circuitry (e.g., as encompassed within a general-purpose processor or other programmable processor) that is temporarily configured by software to perform certain operations. It will be appreciated that the decision to implement a hardware module mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software) may be driven by cost and time considerations.

Accordingly, the term “hardware module” should be understood to encompass a tangible entity that may be physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering embodiments in which hardware modules may be temporarily configured (e.g., programmed), each of the hardware modules need not be configured or instantiated at any one instance in time. For example, where the hardware modules comprise a general-purpose processor configured using software, the general-purpose processor may be configured as respective different hardware modules at different times. Software may accordingly configure a processor, for example, to constitute a particular hardware module at one instance of time and to constitute a different hardware module at a different instance of time.

Hardware modules can provide information to, and receive information from, other hardware modules. Accordingly, the described hardware modules may be regarded as being communicatively coupled. Where multiple of such hardware modules exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) that connect the hard-

ware modules. In some embodiments in which multiple hardware modules are configured or instantiated at different times, communications between such hardware modules may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware modules have access. For example, one hardware module may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware module may then, at a later time, access the memory device to retrieve and process the stored output. Hardware modules may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information).

The various operations of exemplary methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented modules that operate to perform one or more operations or functions. The modules referred to herein may, in some exemplary embodiments, comprise processor-implemented modules.

Similarly, the methods or routines described herein may be at least partially processor-implemented. For example, at least some of the operations of a method may be performed by one or more processors or processor-implemented hardware modules. The performance of certain of the operations may be distributed among the one or more processors, not only residing within a single machine, but deployed across a number of machines. In some example embodiments, the processor or processors may be located in a single location (e.g., within a home environment, an office environment or as a server farm), while in other embodiments the processors may be distributed across a number of geographic locations.

Unless specifically stated otherwise, discussions herein using words such as “processing,” “computing,” “calculating,” “determining,” “presenting,” “displaying,” or the like may refer to actions or processes of a machine (e.g., a computer) that manipulates or transforms data represented as physical (e.g., electronic, magnetic, or optical) quantities within one or more memories (e.g., volatile memory, non-volatile memory, or a combination thereof), registers, or other machine components that receive, store, transmit, or display information.

As used herein any reference to “one embodiment” or “an embodiment” means that a particular element, feature, structure, or characteristic described in connection with the embodiment may be included in at least one embodiment. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment.

Some embodiments may be described using the expression “coupled” and “connected” along with their derivatives. For example, some embodiments may be described using the term “coupled” to indicate that two or more elements are in direct physical or electrical contact. The term “coupled,” however, may also mean that two or more elements are not in direct contact with each other, yet still co-operate or interact with each other. The embodiments are not limited in this context.

As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly

listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

In addition, use of the “a” or “an” is employed to describe elements and components of the embodiments herein. This is done merely for convenience and to give a general sense of the description. This description, and the claims that follow, should be read to include one or at least one and the singular also includes the plural unless it is obvious that it is meant otherwise.

Upon reading this disclosure, those of skill in the art will appreciate still additional alternative structural and functional designs for the approaches described herein. Therefore, while particular embodiments and applications have been illustrated and described, it is to be understood that the disclosed embodiments are not limited to the precise construction and components disclosed herein. Various modifications, changes and variations, which will be apparent to those skilled in the art, may be made in the arrangement, operation and details of the method and apparatus disclosed herein without departing from the spirit and scope defined in the appended claims.

The particular features, structures, or characteristics of any specific embodiment may be combined in any suitable manner and in any suitable combination with one or more other embodiments, including the use of selected features without corresponding use of other features. In addition, many modifications may be made to adapt a particular application, situation or material to the essential scope and spirit of the present invention. It is to be understood that other variations and modifications of the embodiments of the present invention described and illustrated herein are possible in light of the teachings herein and are to be considered part of the spirit and scope of the present invention.

While the preferred embodiments of the invention have been described, it should be understood that the invention is not so limited and modifications may be made without departing from the invention. The scope of the invention is defined by the appended claims, and all devices that come within the meaning of the claims, either literally or by equivalence, are intended to be embraced therein.

It is therefore intended that the foregoing detailed description be regarded as illustrative rather than limiting, and that it be understood that it is the following claims, including all equivalents, that are intended to define the spirit and scope of this invention.

The systems and methods described herein are directed to an improvement to computer functionality and improve the functioning of conventional computers.

What is claimed:

1. A computer system for flood monitoring and insurance provider notification, the computer system comprising:

one or more processors; and

a memory storing executable instructions thereon that, when executed by the one or more processors, cause the one or more processors to:

detect a flood event in a structure, and

transmit information associated with the structure and a prompt for flood reimbursement services to a machine learning (ML) chatbot to cause the ML chatbot to file a flood reimbursement claim with an insurance provider having an insurance policy asso-

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ciated with the structure via telephone by converting a text output into a voice output.

2. The computer system of claim 1, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to perform the detection of the flood event in the structure by receiving a signal from a water sensor.

3. The computer system of claim 1, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to perform the detection of the flood event in the structure by receiving a notification of an occurrence of precipitation exceeding a specified amount.

4. The computer system of claim 1, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to perform the detection of the flood event in the structure by:

receiving a notification of an occurrence of precipitation, and
detecting an interruption of an electricity supply to the structure.

5. The computer system of claim 1, wherein filing the flood reimbursement claim with the insurance provider comprises providing the information associated with the structure to the insurance provider.

6. The computer system of claim 1, wherein the prompt for flood reimbursement services further causes the ML chatbot to request a predetermined payout amount from the insurance provider.

7. The computer system of claim 1, wherein the prompt for flood reimbursement services further causes the ML chatbot to:

collect information from the insurance provider by converting a voice input into a text input, and
communicate the information to a user associated with the structure.

8. A computer-implemented method for flood monitoring and insurance provider notification, the method comprising: detecting a flood event in a structure; and transmitting information associated with the structure and a prompt for flood reimbursement services to a machine learning (ML) chatbot to cause the ML chatbot to file a flood reimbursement claim with an insurance provider having an insurance policy associated with the structure via telephone by converting a text output into a voice output.

9. The computer-implemented method of claim 8, wherein detecting the flood event in the structure comprises receiving a signal from a water sensor.

10. The computer-implemented method of claim 8, wherein detecting of the flood event in the structure comprises receiving a notification of an occurrence of precipitation exceeding a specified amount.

11. The computer-implemented method of claim 8, wherein detecting of the flood event in the structure comprises:

receiving a notification of an occurrence of precipitation, and

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detecting an interruption of an electricity supply to the structure.

12. The computer-implemented method of claim 8, wherein filing the flood reimbursement claim with the insurance provider comprises providing the information associated with the structure to the insurance provider.

13. The computer-implemented method of claim 8, wherein the prompt for flood reimbursement services further causes the ML chatbot to request a predetermined payout amount from the insurance provider.

14. The computer-implemented method of claim 8, wherein the prompt for flood reimbursement services further causes the ML chatbot to:

collect information from the insurance provider by converting a voice input into a text input, and
communicate the information to a user associated with the structure.

15. A computer readable storage medium storing non-transitory computer readable instructions for flood monitoring and insurance provider notification, wherein the instructions when executed on one or more processors cause the one or more processors to:

detect a flood event in a structure, and
transmit information associated with the structure and a prompt for flood reimbursement services to a machine learning (ML) chatbot to cause the ML chatbot to file a flood reimbursement claim with an insurance provider having an insurance policy associated with the structure via telephone by converting a text output into a voice output.

16. The computer readable storage medium of claim 15, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to perform the detection of the flood event in the structure by receiving a signal from a water sensor.

17. The computer readable storage medium of claim 15, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to perform the detection of the flood event in the structure by receiving a notification of an occurrence of precipitation exceeding a specified amount.

18. The computer readable storage medium of claim 15, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to perform the detection of the flood event in the structure by:

receiving a notification of an occurrence of precipitation, and
detecting an interruption of an electricity supply to the structure.

19. The computer readable storage medium of claim 15, wherein filing the flood reimbursement claim with the insurance provider comprises providing the information associated with the structure to the insurance provider.

20. The computer readable storage medium of claim 15, wherein the prompt for flood reimbursement services further causes the ML chatbot to request a predetermined payout amount from the insurance provider.

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